A Pruned Square-Root Unscented Kalman Filter for Bayesian Estimation of Second-Order DSGE Models

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Abstract

We develop a novel filtering methodology—the Pruned Square-Root Unscented Kalman Filter (Pruned SR-UKF)—for likelihood evaluation in Dynamic Stochastic General Equilibrium (DSGE) models solved with second-order perturbation methods. The standard approach augments the state vector to include both first-order states, their Kronecker products, and second-order corrections, yielding an $\mathcal{O}(n_x + n_x^2 + n_x)$-dimensional linear state space amenable to the Kalman filter. This augmentation is computationally infeasible for models with more than approximately 15 state variables, as the per-step cost scales as $\mathcal{O}((n_x + n_x^2)^3)$. Our method instead applies the unscented Kalman filter to the n_x-dimensional first-order state alone, propagates the second-order correction deterministically from the filtered posterior mean, and handles the quadratic observation nonlinearity through sigma-point quadrature. We prove that (i) the sigma-point approximation captures the conditional mean of the quadratic observation equation exactly and its covariance to second order; (ii) the approximation error in the second-order state correction is $\mathcal{O}(\sigma^2)$, where σ denotes the shock standard deviation; and (iii) the per-step cost is $\mathcal{O}(n_x^3)$, yielding a speedup of $\mathcal{O}(n_x^3)$ over the augmented filter. We embed the filter in a Hamiltonian Monte Carlo (HMC) posterior sampler with implicit adjoint gradients through the perturbation solver and validate the methodology on the Schmitt-Grohé and Uribe (2003) small open economy model, demonstrating that the Pruned SR-UKF produces posterior estimates statistically indistinguishable from the augmented Kalman filter at a fraction of the computational cost.

Keywords: DSGE models, second-order perturbation, pruning, unscented Kalman filter, square-root filter, Bayesian estimation, Hamiltonian Monte Carlo.

JEL Classification: C11, C32, C63, E32.

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Contents

1 Introduction

Second-order perturbation methods have become indispensable in macroeconomics. By retaining quadratic terms in the Taylor expansion of the model's policy functions around the deterministic steady state, they capture phenomena that first-order (linear) approximations miss entirely: risk premia, precautionary savings, welfare costs of business cycles, and asymmetric impulse responses (Schmitt-Grohé and Uribe, 2004; Kim et al., 2008). Yet Bayesian estimation of DSGE models solved to second order remains computationally demanding, and the difficulty grows rapidly with model size.

The central bottleneck is likelihood evaluation. The pruned state-space system of Andreasen et al. (2018) decomposes the approximate solution into a first-order stochastic component ${x^f}_t$ and a second-order deterministic correction ${x^s}_t$, then stacks both together with $\mathrm{vec}({x^f}_t {x^f}_t^\top)$ to form an augmented state of dimension $n_x + n_x^2 + n_x$. This augmented system is conditionally linear-Gaussian, so the standard Kalman filter applies. While analytically elegant, the per-step cost of the Kalman filter scales as $\mathcal{O}((2n_x + n_x^2)^3)$, which for a model with $n_x = 30$ state variables implies an augmented dimension of 960 and roughly 8.8×10^8 floating-point operations per time step—repeated for every likelihood evaluation inside a Markov Chain Monte Carlo (MCMC) sampler. This computational barrier has confined second-order Bayesian estimation to small-scale models in practice.

Contribution. We propose the *Pruned Square-Root Unscented Kalman Filter* (Pruned SR-UKF), a filtering methodology that operates on the n_x-dimensional first-order state ${x^f}_t$ alone, while correctly incorporating second-order effects through two mechanisms:

1. The *observation equation* $y_t = g_x({x^f}_t + {x^s}_t) + \frac{1}{2} G_{xx}({x^f}_t \otimes {x^f}_t) + \frac{1}{2} g_{\sigma\sigma} + \varepsilon_t$ is quadratic in ${x^f}_t$. We handle this nonlinearity with the unscented transform (UT), which propagates $2n_x + 1$ sigma points through the observation function and recovers the conditional mean *exactly* and the conditional covariance to second order—because the observation is a polynomial of degree two in ${x^f}_t$.
2. The *second-order correction* ${x^s}_t$ is propagated deterministically from the filtered posterior mean of ${x^f}_t$. We characterize the resulting approximation error and show it is $\mathcal{O}(\sigma^2)$, where σ is the shock standard deviation.

The per-step cost of our filter is $\mathcal{O}(n_x^3)$—a speedup of $\mathcal{O}(n_x^3)$ relative to the augmented Kalman filter. For the $n_x = 7$ Schmitt-Grohé and Uribe (2003) model this represents a factor of approximately 343; for a medium-scale model with $n_x = 30$ the speedup exceeds 10^6.

We embed the Pruned SR-UKF in a complete Bayesian estimation pipeline: maximum a posteriori (MAP) initialization via L-BFGS, Hessian-based mass-matrix preconditioning with eigenvalue capping, and Hamiltonian Monte Carlo (HMC) sampling with implicit adjoint gradients through the perturbation solver (Childers et al., 2024). The

entire pipeline is implemented in TensorFlow with automatic differentiation through the filter recursion, enabling gradient-based posterior exploration without finite-difference approximations.

Related literature. Our work bridges four strands of the literature: second-order perturbation methods (Schmitt-Grohé and Uribe, 2004; Kim et al., 2008; Andreasen et al., 2018), nonlinear filtering for DSGE models (Fernández-Villaverde and Rubio-Ramírez, 2007; Kollmann, 2014; Ivashchenko, 2014; Andreasen, 2012), square-root sigma-point filtering (Julier and Uhlmann, 1997; Wan and van der Merwe, 2000; van der Merwe and Wan, 2001), and gradient-based Bayesian DSGE estimation (Childers et al., 2024). We defer a detailed review to Section 2.

Outline. Section 2 reviews the relevant literature. Section 3 presents the model framework, including the second-order perturbation solution and the pruned state-space system. Section 4 develops the Pruned SR-UKF algorithm in detail. Section 5 states and proves the theoretical properties. Section 6 discusses implementation and numerical considerations. Section 7 addresses potential objections with mathematical rebuttals. Section 8 applies the methodology to the Schmitt-Grohé and Uribe (2003) small open economy model. Section 9 presents Monte Carlo evidence. Section 10 concludes.

2 Literature Review

2.1 Second-Order Perturbation Methods and Pruning

The perturbation approach to solving DSGE models expands the policy functions in a Taylor series around the deterministic steady state. Schmitt-Grohé and Uribe (2004) derive a general second-order perturbation method by reducing the quadratic terms of the equilibrium conditions to a generalized Sylvester equation. Their method applies to any rational-expectations model written in the canonical form $\mathbb{E}_t[H(y_{t+1}, y_t, x_{t+1}, x_t)] = 0$.

A well-known difficulty with second-order (and higher) approximations is that direct simulation of the full nonlinear law of motion can produce explosive sample paths, even when the underlying linear solution is stable. Kim et al. (2008) address this by introducing a "pruning" scheme that decomposes the state into a first-order stochastic component and a second-order correction, and crucially uses only the first-order component in the quadratic terms. This prevents the explosive feedback loop while preserving second-order accuracy.

Andreasen et al. (2018) provide a rigorous analysis of the pruned state-space system for approximations up to third order. They establish stationarity conditions, derive closed-form expressions for unconditional first and second moments, and show how the pruned

system can be cast as a linear state-space model suitable for the Kalman filter—at the cost of augmenting the state vector to include the Kronecker products of the first-order states.

2.2 Nonlinear Filtering for DSGE Estimation

Fernández-Villaverde and Rubio-Ramírez (2007) pioneer likelihood-based Bayesian estimation of nonlinear DSGE models using the particle filter (sequential Monte Carlo). The particle filter provides asymptotically exact likelihood estimates but requires $\mathcal{O}(N_p)$ model evaluations per time step, where N_p is the number of particles—typically 10^3 to 10^5 for acceptable variance.

Kollmann (2014) develops a deterministic alternative by applying the standard Kalman filter to the pruned augmented state space of Kim et al. (2008). This avoids stochastic simulation entirely and produces competitive accuracy relative to particle filters, but inherits the $\mathcal{O}((n_x + n_x^2)^3)$ cost that limits scalability.

Ivashchenko (2014) compares several nonlinear Kalman filters for second-order DSGE estimation, including the Unscented Kalman Filter (UKF), the Central Difference Kalman Filter (CDKF), and the Quadratic Kalman Filter (QKF). He finds the QKF superior in parameter-estimation accuracy but does not employ pruning, instead applying the filters to the full nonlinear dynamics.

Andreasen (2012) applies the CDKF to second- and third-order DSGE solutions, demonstrating good performance with Gaussian, Laplace-distributed, and stochastic volatility shocks, again without the pruning decomposition.

Monfort et al. (2015) develop the Quadratic Kalman Filter for general state-space models with quadratic measurement and transition equations, achieving up to 70% RMSE improvements over the UKF and extended Kalman filter. However, the QKF requires the state-space to be written in a specific quadratic form that may not map directly to the pruned decomposition.

2.2.1 Why Ivashchenko (2014) and Andreasen (2012) Could Avoid Pruning—and Why We Cannot

A natural question arises: if Ivashchenko (2014) successfully applied the UKF, and Andreasen (2012) the CDKF, to second-order DSGE solutions *without* pruning, why does our method require it? The answer involves a fundamental distinction in what these approaches ask the filter to do, and the consequences for stability, theoretical guarantees, and scalability to full Bayesian estimation.

The unpruned approach: sigma points through the full nonlinear dynamics. Both Ivashchenko and Andreasen treat the second-order DSGE solution as a *generic*

nonlinear state-space model:

$$x_{t+1} = h_x\, x_t + \tfrac{1}{2} H_{xx}\,(x_t \otimes x_t) + \tfrac{1}{2} h_{\sigma\sigma} + \eta\, \varepsilon_{t+1}, \tag{1}$$

$$y_t = g_x\, x_t + \tfrac{1}{2} G_{xx}\,(x_t \otimes x_t) + \tfrac{1}{2} g_{\sigma\sigma} + \varepsilon_t^y. \tag{2}$$

The UKF (or CDKF) is then applied as a black-box nonlinear filter: sigma points are propagated through *both* the transition equation (1) and the observation equation (2). The filtered state is the full x_t—there is no decomposition into first- and second-order components.

This approach works in practice for several reasons:

1. **Localized sigma points.** The UKF/CDKF sigma points are constructed around the current filtered mean $\hat{x}_{t|t}$ with spread determined by the filtered covariance $P_{t|t}$. For typical DSGE calibrations with small shocks ($\sigma_\varepsilon \sim 0.01$–$0.1$), the sigma points remain close to the steady state, where the quadratic terms $\frac{1}{2}H_{xx}(x_t \otimes x_t)$ are small relative to the linear term $h_x\, x_t$. The quadratic feedback that causes explosions in *open-loop simulation* is suppressed by the *closed-loop* nature of filtering: each observation corrects the state estimate back toward the true state, preventing the accumulation of quadratic drift.
2. **Point estimation context.** Ivashchenko uses quasi-maximum likelihood (QML) at fixed parameter values; Andreasen likewise uses QML. In both cases, the filter runs at a single parameter configuration (or along a fixed optimization path). If the filter diverges for a particular parameter draw, the likelihood is simply $-\infty$ and that parameter is rejected. Neither author embeds the filter in an MCMC sampler that must evaluate the likelihood at thousands of arbitrary parameter configurations.
3. **Small models.** Both papers use small DSGE models ($n_x \leq 10$), where the quadratic terms are moderate and the probability of filter divergence at reasonable parameter values is low.

Why these reasons are insufficient for Bayesian MCMC estimation. In an MCMC setting—especially with gradient-based samplers like HMC—the filter must evaluate the likelihood at a wide range of parameter configurations, including regions where:

1. $\mathrm{sp}(h_x)$ is close to 1, amplifying the quadratic feedback;
2. $\|H_{xx}\|$ is large relative to the shock variance, making the quadratic terms dominant;
3. the model is near the boundary of Blanchard–Kahn determinacy, where the dynamics are ill-conditioned.

In these regions, the unpruned sigma-point propagation through (1) can produce sigma points that lie far from the steady state. These outlier sigma points generate large $x \otimes x$ terms, which feed back into the next predict step, creating the explosive positive-feedback loop that Kim et al. (2008) identified. Even if such explosions are rare, a single divergent

likelihood evaluation during leapfrog integration can derail an entire HMC trajectory, destroying the sampler's efficiency.

Moreover, the unpruned system (1) is *not stationary*: Andreasen et al. (2018) prove that unconditional moments of the unpruned second-order solution do not exist in general, because the quadratic feedback can generate paths that grow without bound. den Haan and De Wind (2012) further show that the unpruned dynamics can exhibit "spurious oscillations" and "explosive paths" that do not correspond to any feature of the true model solution, but are artifacts of the polynomial approximation operating outside its radius of convergence. Without stationarity, there is no well-defined ergodic distribution, and the filtered state may drift arbitrarily far from the steady state over long sample periods.

Five structural advantages of pruning for our method. The pruned decomposition $x_t = {x^f}_t + {x^s}_t$ resolves all of these issues:

1. **Guaranteed stationarity of ${x^f}_t$.** The first-order state follows linear dynamics ${x^f}_{t+1} = h_x {x^f}_t + \eta\varepsilon_{t+1}$. Under Assumption 2 ($\mathrm{sp}(h_x) < 1$), ${x^f}_t$ is unconditionally stationary with well-defined moments. No quadratic term can cause explosions, regardless of parameter configuration.
2. **Exact predict step.** Because the transition of ${x^f}_t$ is linear, the predict step of our filter is the *exact* Kalman predict—zero approximation error. The UKF/CDKF sigma-point approximation is confined to the observation step, where it handles the *known* quadratic function $\phi(x^f)$. By contrast, the unpruned UKF/CDKF introduces sigma-point approximation error in *both* the predict and update steps, with the predict-step error potentially compounding across time.
3. **Exact conditional mean of the observation.** The observation equation under pruning is *exactly quadratic* in ${x^f}_t$ (Theorem 4.2). The UKF sigma-point mean captures $\mathbb{E}[\phi({x^f}_t) \mid Y_{1:t}]$ with zero error (Theorem 5.1(a)). Under the unpruned approach, the observation is quadratic in the *full* state x_t, but x_t includes second-order terms that are correlated with $x_t \otimes x_t$ in a non-Gaussian way, degrading the sigma-point accuracy.
4. **Bounded ${x^s}_t$ correction.** The second-order correction satisfies ${x^s}_{t+1} = h_x {x^s}_t + \frac{1}{2}H_{xx}(\hat{{x^f}}_{t|t} \otimes \hat{{x^f}}_{t|t}) + \frac{1}{2}h_{\sigma\sigma}$. Since $\mathrm{sp}(h_x) < 1$ and $\hat{{x^f}}_{t|t}$ is bounded (by the filtered covariance bound of Theorem 5.6), the correction ${x^s}_t$ is bounded as well. There is no explosive feedback.
5. **Smooth gradients for HMC.** The log-likelihood under pruning is a smooth function of θ: the linear predict step and the polynomial observation step are both differentiable, and the implicit adjoint gradient through the perturbation solver is well-defined (Childers et al., 2024). Under the unpruned approach, the quadratic *transition* creates gradient instabilities: the derivative of $H_{xx}(x \otimes x)$ with respect

to θ involves products of filtered states with perturbation-solution derivatives, which can amplify numerical errors near the stability boundary.

The key conceptual shift. Ivashchenko and Andreasen use sigma-point filters to *approximate the full nonlinear dynamics*—the filter is a substitute for the intractable optimal nonlinear filter. Our approach is fundamentally different: we use pruning to *separate the problem* into a linear filtering component (${x^f}_t$) and a deterministic propagation component (${x^s}_t$), and use the UKF only to handle the *observation nonlinearity*. This separation is what enables the exact predict step, the exact conditional mean, the stability guarantee, and the smooth gradients—none of which are available in the unpruned approach.

Table 1 summarizes the comparison.

Table 1: Pruned vs. Unpruned Sigma-Point Filtering: Structural Comparison

Property	Unpruned UKF/CDKF	Pruned SR-UKF (this paper)
Filtered state	Full x_t (includes 2nd-order effects)	${x^f}_t$ only (1st-order)
Transition handled by	Sigma-point propagation (approximate)	Exact Kalman predict (exact)
Observation handled by	Sigma-point propagation (approximate)	Sigma-point propagation (exact mean for quadratic)
2nd-order correction	Implicit in filtered x_t	Explicit deterministic ${x^s}_t$ update
Stationarity	Not guaranteed (moments may not exist)	Guaranteed under $\mathrm{sp}(h_x) < 1$
Stability under MCMC	Can diverge at extreme θ	Bounded for all θ satisfying BK
Predict-step error	$\mathcal{O}(\lVert P \rVert^{3/2})$ (sigma-point approx.)	Zero (exact linear predict)
Observation-mean error	$\mathcal{O}(\lVert P \rVert^{3/2})$ for quadratic in x_t	Zero (exact for quadratic in ${x^f}_t$)
Gradient smoothness	Unstable near $\mathrm{sp}(h_x) \to 1$	Smooth for all $\mathrm{sp}(h_x) < 1$

We return to the question of why pruning is essential—and rebut it in detail for skeptical reviewers—in Section 7.3.

Gap in the literature. No existing paper combines the sigma-point methodology of the UKF with the pruned state-space decomposition of Andreasen et al. (2018). That is, no prior work applies the UKF to the first-order state ${x^f}_t$ alone while propagating the second-order correction ${x^s}_t$ deterministically, using sigma-point quadrature to capture the quadratic observation nonlinearity. Table 2 summarizes the positioning relative to all filtering approaches.

Table 2: Comparison of Filtering Approaches for Second-Order DSGE Models

Method	State dim.	Filter type	Stability	Per-step cost
Augmented KF	$n_x+n_x^2+n_x$	Linear KF	Pruning	$\mathcal{O}((2n_x+n_x^2)^3)$
UKF (no pruning)	n_x	Sigma-point	None	$\mathcal{O}(n_x^3)$
Particle filter	n_x	Monte Carlo	N/A	$\mathcal{O}(N_p n_x^2)$
Pruned SR-UKF	$\boldsymbol{n_x}$	**Sigma-point**	**Pruning**	$\boldsymbol{\mathcal{O}(n_x^3)}$

2.3 Square-Root Sigma-Point Filtering

The Unscented Transform (UT) of Julier and Uhlmann (1997) propagates a symmetric set of deterministically chosen "sigma points" through a nonlinear function to recover the posterior mean and covariance without linearization. Wan and van der Merwe (2000) formalize the Unscented Kalman Filter (UKF) and demonstrate that it captures the posterior mean and covariance correctly to third order for Gaussian inputs and arbitrary nonlinearities—and *exactly* for polynomial nonlinearities of degree two.

van der Merwe and Wan (2001) introduce the Square-Root UKF (SR-UKF), which propagates the Cholesky factor of the covariance matrix directly using QR decompositions and rank-one Cholesky updates/downdates. The SR-UKF guarantees positive semi-definiteness of the covariance at every step (assuming non-negative sigma-point weights), roughly halves the numerical condition number, and is no more expensive than the standard UKF.

2.4 Gradient-Based Bayesian DSGE Estimation

Childers et al. (2024) develop a framework for differentiable state-space models, combining implicit differentiation through the perturbation solver with Hamiltonian Monte Carlo sampling. Their approach avoids differentiating through the QZ decomposition and Sylvester equation by instead differentiating the *implicit equilibrium conditions* that define the perturbation solution—a technique known as the adjoint or implicit function theorem method. This provides exact gradients of the log-posterior with respect to structural parameters at $\mathcal{O}(n^2)$ cost per gradient evaluation, enabling the use of HMC and the No-U-Turn Sampler (NUTS) for Bayesian DSGE estimation.

3 Model Framework

3.1 General DSGE in Expectations Form

Consider a DSGE model with n_y control (jump) variables $y_t \in \mathbb{R}^{n_y}$ and n_x state (predetermined) variables $x_t \in \mathbb{R}^{n_x}$, subject to n_ε structural shocks $\varepsilon_t \sim \mathcal{N}(0, I_{n_\varepsilon})$. The

equilibrium conditions are

$$\mathbb{E}_t[H(y_{t+1},\, y_t,\, x_{t+1},\, x_t)] = 0, \tag{3}$$

where $H : \mathbb{R}^{n_y} \times \mathbb{R}^{n_y} \times \mathbb{R}^{n_x} \times \mathbb{R}^{n_x} \to \mathbb{R}^{n_y+n_x}$ collects the $n_y + n_x$ equilibrium equations, and the expectation is conditional on information at time t. The state transition includes the shock: $x_{t+1} = \tilde{h}(y_t, x_t) + \eta(\theta)\, \varepsilon_{t+1}$, where $\eta \in \mathbb{R}^{n_x \times n_\varepsilon}$ is the shock-impact matrix.

Let $(\bar{y}, \bar{x})$ denote the deterministic steady state satisfying $H(\bar{y}, \bar{y}, \bar{x}, \bar{x}) = 0$. We work with deviations from steady state: $\hat{y}_t \equiv y_t - \bar{y}$, $\hat{x}_t \equiv x_t - \bar{x}$.

3.2 First-Order Perturbation Solution

The first-order (linear) approximation yields policy and transition functions

$$\hat{y}_t = g_x\, \hat{x}_t, \tag{4}$$

$$\hat{x}_{t+1} = h_x\, \hat{x}_t + \eta\, \varepsilon_{t+1}, \tag{5}$$

where $g_x \in \mathbb{R}^{n_y \times n_x}$ and $h_x \in \mathbb{R}^{n_x \times n_x}$ are determined by the first-order conditions. Define the total-variable vector $v = (y', y, x', x) \in \mathbb{R}^{2(n_y+n_x)}$ and the Jacobians $H_{y'}, H_y, H_{x'}, H_x$ evaluated at the steady state. The first-order conditions are:

$$H_x + H_y\, g_x + (H_{x'} + H_{y'}\, g_x)\, h_x = 0, \tag{6}$$

together with the Blanchard–Kahn condition requiring the number of stable generalized eigenvalues of the pencil $(H_{x'} + H_{y'}, -(H_x + H_y))$ to equal n_x. The matrices g_x, h_x are typically recovered via the QZ (generalized Schur) decomposition (Klein, 2000).

Assumption 1. *The Blanchard–Kahn condition holds: the generalized Schur decomposition of the pencil* (Γ_1, Γ_0) *with* $\Gamma_0 = [H_{x'} \mid H_{y'}]$ *and* $\Gamma_1 = [-H_x \mid -H_y]$ *yields exactly* n_x *stable generalized eigenvalues (with modulus strictly less than* $1 + \delta$ *for some small* $\delta > 0$*).*

Assumption 2. *The spectral radius of* h_x *satisfies* $\mathrm{sp}(h_x) \equiv \max_i |\lambda_i(h_x)| < 1$*, ensuring asymptotic stability of the first-order state dynamics.*

3.3 Second-Order Perturbation Solution

Following Schmitt-Grohé and Uribe (2004), the second-order approximation adds quadratic correction terms:

$$\hat{y}_t = g_x\, \hat{x}_t + \tfrac{1}{2} G_{xx}\, (\hat{x}_t \otimes \hat{x}_t) + \tfrac{1}{2} g_{\sigma\sigma}\, \sigma^2, \tag{7}$$

$$\hat{x}_{t+1} = h_x\, \hat{x}_t + \eta\, \varepsilon_{t+1} + \tfrac{1}{2} H_{xx}\, (\hat{x}_t \otimes \hat{x}_t) + \tfrac{1}{2} h_{\sigma\sigma}\, \sigma^2, \tag{8}$$

where $G_{xx} \in \mathbb{R}^{n_y \times n_x^2}$, $H_{xx} \in \mathbb{R}^{n_x \times n_x^2}$, $g_{\sigma\sigma} \in \mathbb{R}^{n_y}$, and $h_{\sigma\sigma} \in \mathbb{R}^{n_x}$ are the second-order coefficient matrices. Throughout, we set $\sigma = 1$ (full stochasticity) so that $g_{\sigma\sigma}$ and $h_{\sigma\sigma}$ are vectors.

The Sylvester equation for G_{xx}, H_{xx}. Define $n \equiv n_y + n_x$, $M \equiv [g_x h_x;\ g_x;\ h_x;\ I_{n_x}] \in \mathbb{R}^{2n \times n_x}$, and the Hessian tensor $\Psi \in \mathbb{R}^{n \times 2n \times 2n}$ where $\Psi_{ijk} = \partial^2 H_i / (\partial v_j\, \partial v_k)$ evaluated at steady state. The stacked second-order coefficient $X \equiv [G_{xx};\ H_{xx}] \in \mathbb{R}^{n \times n_x^2}$ solves the generalized Sylvester equation

$$A\, X\, (h_x \otimes h_x)^\top + D\, X = -C, \tag{9}$$

where $A = [H_{y'} \quad 0_{n_y \times n_x};\ 0_{n_x \times n_y} \quad 0_{n_x}]$, $D = [H_y \quad H_{y'} g_x + H_{x'}]$, and $C \in \mathbb{R}^{n \times n_x^2}$ is formed from the quadratic terms $C_{i\cdot} = \mathrm{vec}(M^\top \Psi_i M)^\top$ for each equation i. The Sylvester equation (9) is solved efficiently by the Bartels–Stewart algorithm: Schur decomposition of h_x gives $h_x = QTQ^\top$, which induces a block-triangular structure in $h_x \otimes h_x$, reducing the problem to a sequence of $n \times n$ linear systems (Bartels and Stewart, 1972).

The volatility corrections $g_{\sigma\sigma}, h_{\sigma\sigma}$. The constants $g_{\sigma\sigma}, h_{\sigma\sigma}$ solve a linear system

$$\begin{bmatrix} H_y & H_{y'} g_x + H_{x'} \end{bmatrix} \begin{bmatrix} g_{\sigma\sigma} \\ h_{\sigma\sigma} \end{bmatrix} = -B, \tag{10}$$

where $B \in \mathbb{R}^n$ depends on η, the Hessians Ψ, and the second-order coefficients G_{xx}, H_{xx} already obtained.

3.4 The Pruned State-Space System

Direct simulation of (8) feeds $\hat{x}_t \otimes \hat{x}_t$ (including second-order terms) back into the quadratic dynamics, creating a feedback loop that can generate explosive paths even when $\mathrm{sp}(h_x) < 1$. Kim et al. (2008) and Andreasen et al. (2018) resolve this by decomposing the state:

Definition 3.1 (Pruned state-space system)**.** *Define the first-order state ${x^f}_t$ and the second-order correction ${x^s}_t$ by:*

$$\begin{aligned} {x^f}_{t+1} &= h_x\, {x^f}_t + \eta\, \varepsilon_{t+1}, && (11)\\ {x^s}_{t+1} &= h_x\, {x^s}_t + \tfrac{1}{2} H_{xx}\, ({x^f}_t \otimes {x^f}_t) + \tfrac{1}{2} h_{\sigma\sigma}, && (12)\\ y_t &= g_x\, ({x^f}_t + {x^s}_t) + \tfrac{1}{2} G_{xx}\, ({x^f}_t \otimes {x^f}_t) + \tfrac{1}{2} g_{\sigma\sigma} + \varepsilon_t^y, && (13) \end{aligned}$$

where $\varepsilon_t^y \sim \mathcal{N}(0, R)$ is measurement noise.

The key feature of pruning is that *only the first-order state* ${x^f}_t$ *enters the quadratic terms.* Since ${x^f}_t$ follows linear-Gaussian dynamics (11), the Kronecker product ${x^f}_t \otimes {x^f}_t$ remains bounded in expectation, preventing explosive behavior.

3.5 The Augmented State-Space (Existing Approach)

Andreasen et al. (2018) and Kollmann (2014) define the augmented state

$$\xi_t \equiv \begin{bmatrix} {x^f}_t \\ \mathrm{vec}({x^f}_t {x^f}_t^\top) \\ {x^s}_t \end{bmatrix} \in \mathbb{R}^{n_x + n_x^2 + n_x} \tag{14}$$

and show that the system (11)–(13) can be written as a linear state-space model:

$$\xi_{t+1} = \mathcal{T}\,\xi_t + c + \nu_{t+1}, \quad \mathbb{E}[\nu_t \nu_t^\top] = \mathcal{Q}, \tag{15}$$

$$y_t = \mathcal{Z}\,\xi_t + d + \varepsilon_t^y, \tag{16}$$

where

$$\mathcal{T} = \begin{bmatrix} h_x & 0 & 0 \\ 0 & h_x \otimes h_x & 0 \\ 0 & \frac{1}{2} H_{xx} & h_x \end{bmatrix}, \quad c = \begin{bmatrix} 0 \\ \mathrm{vec}(\eta\eta^\top) \\ \frac{1}{2} h_{\sigma\sigma} \end{bmatrix}, \tag{17}$$

and $\mathcal{Z}$ and d select and combine the relevant rows to form the observation equation. The process-noise covariance $\mathcal{Q}$ is block-diagonal with the $(1,1)$ block equal to $\eta\eta^\top$ and the $(2,2)$ block $\mathcal{Q}_{kk}$ given by the *unconditional* covariance of the Kronecker innovations (Andreasen et al., 2018, eq. 16):

$$\mathcal{Q}_{kk} = (\Sigma_a \otimes \Sigma_b) + (\Sigma_b \otimes \Sigma_a) + (P_1 \otimes P_1) + \text{cross terms}, \tag{18}$$

where $\Sigma_a = h_x P_f h_x^\top$, $\Sigma_b = \eta\eta^\top$, $P_1 = \eta\eta^\top$, and P_f is the unconditional covariance of ${x^f}_t$ solving the discrete Lyapunov equation $P_f = h_x P_f h_x^\top + \eta\eta^\top$.

The standard Kalman filter applied to the augmented system (15)–(16) yields an exact (under the Gaussian approximation) log-likelihood, but the per-step cost is

$$\mathcal{O}\Big((n_x + n_x^2 + n_x)^3\Big) = \mathcal{O}(n_x^6), \tag{19}$$

since the dominant operations (Kalman gain, covariance update) involve matrices of dimension $2n_x + n_x^2$.

Remark 3.2. *For the Schmitt-Grohé and Uribe (2003) model with $n_x = 7$, the augmented dimension is $7+49+7 = 63$. The Kalman filter operates on 63×63 matrices—manageable but expensive when embedded in an MCMC sampler requiring 10^4–10^5 likelihood evalua-*

tions. For a medium-scale New Keynesian model with $n_x = 30$, the augmented dimension is 960, making the approach infeasible.

4 The Pruned Square-Root Unscented Kalman Filter

4.1 Overview and Intuition

The central idea of the Pruned SR-UKF is to exploit the structure of the pruned system (11)–(13):

1. The *transition* of ${x^f}_t$ is *linear*: ${x^f}_{t+1} = h_x\, {x^f}_t + \eta\, \varepsilon_{t+1}$. No sigma-point propagation is needed for the predict step—the Kalman filter predict equations apply exactly.
2. The *observation* equation is *quadratic* in ${x^f}_t$ (with ${x^s}_t$ entering linearly as a known offset). The unscented transform handles this nonlinearity by propagating $2n_x + 1$ sigma points through the observation function.
3. The *second-order correction* ${x^s}_t$ depends on the Kronecker product ${x^f}_t \otimes {x^f}_t$, which under Gaussian filtering has conditional expectation $\mathbb{E}[{x^f}_t \otimes {x^f}_t \mid Y_{1:t}] = \hat{x^f}_{t|t} \otimes \hat{x^f}_{t|t} + \mathrm{vec}(P_{t|t})$. We approximate this by $\hat{x^f}_{t|t} \otimes \hat{x^f}_{t|t}$ alone—the "certainty equivalence" approximation—incurring an $\mathcal{O}(\sigma^2)$ error that we characterize in Section 5.

The filter state is $(\hat{x^f}_{t|t}, S_{t|t}, {x^s}_t)$, where $S_{t|t}$ is the Cholesky factor of the filtered covariance $P_{t|t} = S_{t|t} S_{t|t}^\top$. The dimension of the covariance is $n_x \times n_x$, not $(2n_x + n_x^2) \times (2n_x + n_x^2)$.

4.2 Sigma-Point Construction

Let $L \equiv n_x$ be the state dimension. Choose tuning parameters $\alpha > 0$, $\beta \geq 0$, $\kappa \geq 0$, and define

$$\lambda = \alpha^2(L + \kappa) - L, \qquad \gamma = \sqrt{L + \lambda}. \tag{20}$$

Given the state estimate $\hat{x} \in \mathbb{R}^L$ and its Cholesky factor $S \in \mathbb{R}^{L \times L}$ (so $P = SS^\top$), define $2L + 1$ sigma points:

$$\chi^{(0)} = \hat{x}, \tag{21}$$

$$\chi^{(i)} = \hat{x} + \gamma\, S_{:,i}, \quad i = 1, \ldots, L, \tag{22}$$

$$\chi^{(L+i)} = \hat{x} - \gamma\, S_{:,i}, \quad i = 1, \ldots, L, \tag{23}$$

where $S_{:,i}$ denotes the i-th column of S. The associated weights are:

$$w_m^{(0)} = \frac{\lambda}{L+\lambda}, \qquad w_c^{(0)} = w_m^{(0)} + (1-\alpha^2+\beta), \tag{24}$$

$$w_m^{(j)} = w_c^{(j)} = \frac{1}{2(L+\lambda)}, \qquad j = 1, \ldots, 2L. \tag{25}$$

We use the default parameters $\alpha = 1$, $\beta = 2$, $\kappa = \max(0, 3-L)$, which ensure $w_c^{(0)} \geq 0$ for all $L \geq 1$ and provide exact recovery of the Gaussian mean and covariance for affine transformations.

4.3 Initialization

The filter is initialized at the unconditional moments of the first-order state:

$$\hat{x^f}_{0|0} = 0, \qquad P_{0|0} = P_f + \epsilon\, I_{n_x}, \qquad {x^s}_0 = 0, \tag{26}$$

where P_f solves the discrete Lyapunov equation

$$P_f = h_x\, P_f\, h_x^\top + \eta\eta^\top \tag{27}$$

and $\epsilon > 0$ is a small nugget (e.g., 10^{-10}) ensuring positive definiteness.

The Lyapunov equation is solved iteratively: starting from $P^{(0)} = \eta\eta^\top$, iterate $P^{(k+1)} = h_x P^{(k)} h_x^\top + \eta\eta^\top$ until $\|P^{(k+1)} - P^{(k)}\|_F / \|P^{(k+1)}\|_F < \tau$ with tolerance $\tau = 10^{-12}$.

4.4 Predict Step

Since the transition ${x^f}_{t+1} = h_x\, {x^f}_t + \eta\, \varepsilon_{t+1}$ is linear, the predict step is exact:

$$\hat{x^f}_{t+1|t} = h_x\, \hat{x^f}_{t|t}, \tag{28}$$

$$P_{t+1|t} = h_x\, P_{t|t}\, h_x^\top + Q_f, \tag{29}$$

where $Q_f = \eta\eta^\top + \epsilon\, I_{n_x}$.

In square-root form, we compute $S_{t+1|t}$ from the QR decomposition of the matrix

$$\mathcal{R}_p = \begin{bmatrix} \sqrt{w_c^{(0)}}\, (h_x\, S_{t|t})^\top \\ \vdots \\ S_{Q_f}^\top \end{bmatrix}, \tag{30}$$

where $S_{Q_f} = \mathrm{chol}(Q_f)$, so that $S_{t+1|t}$ is the R factor of the QR decomposition (appropriately transposed and sign-corrected).

Remark 4.1. *Because the transition is linear, the sigma-point propagation* $\chi_+^{(j)} = h_x\, \chi^{(j)}$

and the weighted-mean formula $\sum_j w_m^{(j)} \chi_+^{(j)} = h_x \hat{x}$ are exactly the Kalman predict. The QR form is used for numerical stability, not for handling nonlinearity.

4.5 Observation Update

4.5.1 Sigma-point propagation through the observation function.

Generate fresh sigma points $\{\tilde{\chi}^{(j)}\}_{j=0}^{2L}$ from $(\hat{x^f}_{t+1|t},\ S_{t+1|t})$ using (21)–(23). Define the observation function

$$\phi(x) \equiv g_{\text{obs}}\,(x + {x^s}_t) + \tfrac{1}{2}\,G_{\text{obs}}\,(x \otimes x) + \tfrac{1}{2}\,g_{\text{obs},\sigma\sigma}, \tag{31}$$

where $g_{\text{obs}} \in \mathbb{R}^{n_y \times n_x}$, $G_{\text{obs}} \in \mathbb{R}^{n_y \times n_x^2}$, and $g_{\text{obs},\sigma\sigma} \in \mathbb{R}^{n_y}$ are the observation-mapped rows of g_x, G_{xx}, and $g_{\sigma\sigma}$ (selecting observed controls and states). Propagate:

$$z^{(j)} = \phi(\tilde{\chi}^{(j)}), \quad j = 0, 1, \ldots, 2L. \tag{32}$$

Remark 4.2. *The function ϕ is a polynomial of degree two in its argument x: it is affine in x (through $g_{obs}(x + {x^s}_t)$) plus quadratic (through $G_{obs}(x \otimes x)$). The second-order correction ${x^s}_t$ enters only linearly, as a known shift.*

4.5.2 Innovation statistics.

Predicted measurement mean:

$$\bar{z}_{t+1} = \sum_{j=0}^{2L} w_m^{(j)}\, z^{(j)}. \tag{33}$$

Innovation:

$$v_{t+1} = y_{t+1} - \bar{z}_{t+1}. \tag{34}$$

Innovation covariance (square-root form): compute S_{yy} from the QR decomposition of

$$\mathcal{R}_y = \begin{bmatrix} \sqrt{w_c^{(0)}}\,(z^{(0)} - \bar{z}_{t+1})^\top \\ \sqrt{w_c^{(1)}}\,(z^{(1)} - \bar{z}_{t+1})^\top \\ \vdots \\ \sqrt{w_c^{(2L)}}\,(z^{(2L)} - \bar{z}_{t+1})^\top \\ S_R^\top \end{bmatrix}, \tag{35}$$

where $S_R = \text{chol}(R)$ is the Cholesky factor of the measurement noise. Then $P_{yy} = S_{yy} S_{yy}^\top$.

4.5.3 Cross-covariance and Kalman gain.

The state–measurement cross-covariance is

$$P_{xz} = \sum_{j=0}^{2L} w_c^{(j)} \, (\tilde{\chi}^{(j)} - \hat{x^f}_{t+1|t}) \, (z^{(j)} - \bar{z}_{t+1})^\top. \tag{36}$$

The Kalman gain is

$$K = P_{xz} \, P_{yy}^{-1} = P_{xz} \, S_{yy}^{-\top} \, S_{yy}^{-1}, \tag{37}$$

computed via two triangular solves.

4.5.4 Filtered state and covariance.

The filtered mean is

$$\hat{x^f}_{t+1|t+1} = \hat{x^f}_{t+1|t} + K \, v_{t+1}. \tag{38}$$

The filtered covariance is updated as

$$P_{t+1|t+1} = P_{t+1|t} - K \, P_{yy} \, K^\top + \epsilon \, I_{n_x}, \tag{39}$$

symmetrized via $P \leftarrow \frac{1}{2}(P + P^\top)$, and the Cholesky factor $S_{t+1|t+1} = \mathrm{chol}(P_{t+1|t+1})$ is recomputed. The nugget $\epsilon > 0$ ensures positive definiteness.

Remark 4.3 (On the covariance update form)**.** *The classical SR-UKF of van der Merwe and Wan (2001) uses rank-one Cholesky downdates to maintain the square-root factor directly: S_u is obtained by successively downdating S_p with the columns of $K \, S_{yy}$. While this guarantees PSD preservation under exact arithmetic and non-negative weights, the rank-one downdate can fail in finite precision when $K P_{yy} K^\top$ approaches $P_{t+1|t}$ (near-degenerate innovation). We adopt the computationally equivalent approach of* (39) *with nugget regularization, which is robust in automatic-differentiation frameworks where Cholesky downdates are not readily available. See Section 7.4 for a detailed discussion.*

4.6 Deterministic x^s Propagation

After computing the filtered mean $\hat{x^f}_{t+1|t+1}$, the second-order correction is updated deterministically:

$$x^s{}_{t+1} = h_x \, x^s{}_t + \tfrac{1}{2} H_{xx} \, (\hat{x^f}_{t+1|t+1} \otimes \hat{x^f}_{t+1|t+1}) + \tfrac{1}{2} h_{\sigma\sigma}. \tag{40}$$

This equation uses the *filtered* (posterior) mean of $x^f{}_{t+1}$ in the Kronecker product, rather than the conditional expectation $\mathbb{E}[x^f{}_{t+1} \otimes x^f{}_{t+1} \mid Y_{1:t+1}]$. The latter equals $\hat{x^f}_{t+1|t+1} \otimes \hat{x^f}_{t+1|t+1} + \mathrm{vec}(P_{t+1|t+1})$; the omitted term $\frac{1}{2} H_{xx} \, \mathrm{vec}(P_{t+1|t+1})$ constitutes the approximation bias, which we analyze in Section 5.2.

4.7 Log-Likelihood

The Gaussian log-likelihood contribution at time $t+1$ is

$$\ell_{t+1} = -\frac{n_y}{2}\ln(2\pi) - \ln|\det(S_{yy})| - \frac{1}{2}\|S_{yy}^{-1}\, v_{t+1}\|^2, \tag{41}$$

where $\ln|\det(S_{yy})| = \sum_{i=1}^{n_y} \ln|S_{yy,ii}|$. The total log-likelihood is $\ell(Y_{1:T} \mid \theta) = \sum_{t=1}^{T} \ell_t$.

4.8 Complete Algorithm

Algorithm 1 summarizes the Pruned SR-UKF.

Algorithm 1 Pruned Square-Root UKF for Second-Order DSGE

Require: Model solution $(g_x, h_x, \eta, G_{xx}, H_{xx}, g_{\sigma\sigma}, h_{\sigma\sigma})$; observation matrices $(g_{\text{obs}}, G_{\text{obs}}, g_{\text{obs},\sigma\sigma})$; data $Y = \{y_1, \ldots, y_T\}$; measurement noise R.

Ensure: Log-likelihood ℓ.

Solve Lyapunov equation for P_f; set $S_0 \leftarrow \text{chol}(P_f + \epsilon I)$, $\hat{x^f}_0 \leftarrow 0$, $x^s{}_0 \leftarrow 0$, $\ell \leftarrow 0$.
for $t = 0, 1, \ldots, T-1$ **do**
  Predict: $\hat{x^f}^{-} \leftarrow h_x\, \hat{x^f}_t$; $S^- \leftarrow \text{QR-predict}(h_x, S_t, S_{Q_f})$.
  Sigma points: Generate $\{\tilde{\chi}^{(j)}\}$ from $(\hat{x^f}^{-}, S^-)$.
  Observation propagation: $z^{(j)} \leftarrow \phi(\tilde{\chi}^{(j)})$ for each j.
  Innovation: $\bar{z} \leftarrow \sum w_m^{(j)} z^{(j)}$; $v \leftarrow y_{t+1} - \bar{z}$.
  Innovation covariance: $S_{yy} \leftarrow \text{QR-obs}(\{z^{(j)}\}, \bar{z}, S_R)$.
  Cross-covariance: $P_{xz} \leftarrow \sum w_c^{(j)} (\tilde{\chi}^{(j)} - \hat{x^f}^{-})(z^{(j)} - \bar{z})^\top$.
  Kalman gain: $K \leftarrow P_{xz}\,(S_{yy} S_{yy}^\top)^{-1}$.
  Filter update: $\hat{x^f}_{t+1} \leftarrow \hat{x^f}^{-} + Kv$; $P^+ \leftarrow S^- S^{-\top} - K S_{yy} S_{yy}{}^\top K^\top + \epsilon I$; $S_{t+1} \leftarrow \text{chol}(\frac{1}{2}(P^+ + P^{+\top}))$.
  x^s **update:** $x^s{}_{t+1} \leftarrow h_x\, x^s{}_t + \frac{1}{2} H_{xx}(\hat{x^f}_{t+1} \otimes \hat{x^f}_{t+1}) + \frac{1}{2} h_{\sigma\sigma}$.
  Log-likelihood: $\ell \leftarrow \ell - \frac{n_y}{2}\ln(2\pi) - \sum_i \ln|S_{yy,ii}| - \frac{1}{2}\|S_{yy}^{-1} v\|^2$.
end for
return ℓ.

5 Theoretical Properties

In this section we establish the key analytical properties of the Pruned SR-UKF: moment matching of the observation equation (Theorem 5.1), characterization of the x^s approximation bias (Theorem 5.2), computational complexity (Theorem 5.5), stability (Theorem 5.6), and two limiting-case theorems (Theorems 5.7 and 5.8).

5.1 Moment Matching for the Quadratic Observation

Proposition 5.1 (Exact mean, second-order covariance)**.** *Let $x^f \sim \mathcal{N}(\hat{x}, P)$ with $P = SS^\top$, and define the quadratic observation function*

$$\phi(x) = g_{\text{obs}}(x + x^s) + \tfrac{1}{2}\, G_{\text{obs}}\,(x \otimes x) + \tfrac{1}{2}\, g_{\text{obs},\sigma\sigma}, \tag{42}$$

where $x^s \in \mathbb{R}^{n_x}$ is a fixed offset. Then the sigma-point approximations satisfy:

(a) The predicted measurement mean $\bar{z} = \sum_{j=0}^{2L} w_m^{(j)}\, \phi(\chi^{(j)})$ equals $\mathbb{E}[\phi(x^f)]$ exactly:

$$\bar{z} = g_{\text{obs}}(\hat{x} + x^s) + \tfrac{1}{2}\, G_{\text{obs}}\,(\hat{x} \otimes \hat{x} + \text{vec}(P)) + \tfrac{1}{2}\, g_{\text{obs},\sigma\sigma}. \tag{43}$$

(b) The predicted measurement covariance $P_{zz} = \sum_{j=0}^{2L} w_c^{(j)}\,(z^{(j)} - \bar{z})(z^{(j)} - \bar{z})^\top$ agrees with $\text{Var}[\phi(x^f)]$ to second order in P:

$$P_{zz} = \text{Var}[\phi(x^f)] + \mathcal{O}(\|P\|^2). \tag{44}$$

Proof. **Part (a).** Write $\phi(x) = a + B\,x + \frac{1}{2}\, G_{\text{obs}}\,(x \otimes x)$, where $a = g_{\text{obs}}\, x^s + \frac{1}{2}\, g_{\text{obs},\sigma\sigma}$ and $B = g_{\text{obs}}$.

The expectation is

$$\mathbb{E}[\phi(x^f)] = a + B\,\hat{x} + \tfrac{1}{2}\, G_{\text{obs}}\, \mathbb{E}[x^f \otimes x^f]. \tag{45}$$

For $x^f \sim \mathcal{N}(\hat{x}, P)$, we have $\mathbb{E}[x^f \otimes x^f] = \hat{x} \otimes \hat{x} + \text{vec}(P)$, giving (43).

Now consider the sigma-point mean. By the symmetry of the sigma points, $\sum_j w_m^{(j)} \chi^{(j)} = \hat{x}$ (the UT preserves the mean for any function). For the quadratic term, using $\chi^{(i)} = \hat{x} + \gamma\, S_{:,i}$ and $\chi^{(L+i)} = \hat{x} - \gamma\, S_{:,i}$:

$$\begin{aligned}
&\sum_{j=0}^{2L} w_m^{(j)}\,(\chi^{(j)} \otimes \chi^{(j)}) \\
&\quad = w_m^{(0)}\,(\hat{x} \otimes \hat{x}) + \sum_{i=1}^{L} w_m^{(i)} \Big[(\hat{x} + \gamma S_{:,i}) \otimes (\hat{x} + \gamma S_{:,i}) + (\hat{x} - \gamma S_{:,i}) \otimes (\hat{x} - \gamma S_{:,i})\Big] \\
&\quad = w_m^{(0)}\,(\hat{x} \otimes \hat{x}) + \sum_{i=1}^{L} w_m^{(i)} \Big[2(\hat{x} \otimes \hat{x}) + 2\gamma^2\,(S_{:,i} \otimes S_{:,i})\Big] \\
&\quad = \Big(w_m^{(0)} + 2L\, w_m^{(1)}\Big)(\hat{x} \otimes \hat{x}) + 2\gamma^2 w_m^{(1)} \sum_{i=1}^{L} (S_{:,i} \otimes S_{:,i}).
\end{aligned} \tag{46}$$

From (24)–(25): $w_m^{(0)} + 2L\, w_m^{(1)} = \frac{\lambda}{L+\lambda} + \frac{2L}{2(L+\lambda)} = 1$. Also, $2\gamma^2 w_m^{(1)} = 2(L+\lambda) \cdot \frac{1}{2(L+\lambda)} = 1$,

and $\sum_{i=1}^{L} S_{:,i} \otimes S_{:,i} = \mathrm{vec}(SS^\top) = \mathrm{vec}(P)$. Therefore

$$\sum_{j=0}^{2L} w_m^{(j)} \left(\chi^{(j)} \otimes \chi^{(j)}\right) = \hat{x} \otimes \hat{x} + \mathrm{vec}(P), \tag{47}$$

and $\bar{z} = a + B\hat{x} + \frac{1}{2} G_{\mathrm{obs}}(\hat{x} \otimes \hat{x} + \mathrm{vec}(P)) = \mathbb{E}[\phi(x^f)]$. □ (Part a)

Part (b). The true variance of $\phi(x^f) = a + Bx^f + \frac{1}{2} G_{\mathrm{obs}}(x^f \otimes x^f)$ involves moments of x^f up to fourth order. For Gaussian x^f, the fourth central moments are determined by P via $\mathbb{E}[(\Delta x)^{\otimes 4}] = \mathrm{vec}(P)\mathrm{vec}(P)^\top + (I + K_{n_x,n_x})(P \otimes P)$, where K_{n_x,n_x} is the commutation matrix.

The sigma-point covariance captures the leading terms (those linear in P) exactly. The error arises from the fourth-order terms, which contribute $\mathcal{O}(\|P\|^2)$. Specifically, the UKF covariance approximation for a degree-d polynomial is exact through order $2d - 1$ in the standard deviations when $\alpha = 1$ and $\kappa = 3 - L$ (Julier and Uhlmann, 2004). For $d = 2$, this means exactness through third order, implying the error in the covariance is fourth order in the standard deviations, or equivalently $\mathcal{O}(\|P\|^2)$. □

5.2 Bias Characterization of the x^s Update

Proposition 5.2 (x^s approximation bias)**.** *Let $\hat{x^f}_{t|t}$ and $P_{t|t}$ be the filtered mean and covariance of ${x^f}_t$ given observations $Y_{1:t}$, and let ${x^s}_t^\star$ denote the Bayesian optimal second-order correction obtained by using $\mathbb{E}[{x^f}_t \otimes {x^f}_t \mid Y_{1:t}] = \hat{x^f}_{t|t} \otimes \hat{x^f}_{t|t} + \mathrm{vec}(P_{t|t})$ in (12). The Pruned SR-UKF correction ${x^s}_{t+1}^{\mathrm{UKF}}$ from (40) satisfies*

$${x^s}_{t+1}^\star - {x^s}_{t+1}^{\mathrm{UKF}} = \tfrac{1}{2} H_{xx} \,\mathrm{vec}(P_{t|t}) + h_x \left({x^s}_t^\star - {x^s}_t^{\mathrm{UKF}}\right). \tag{48}$$

In particular, the per-step bias increment is $\Delta_t \equiv \frac{1}{2} H_{xx} \,\mathrm{vec}(P_{t|t})$, and the cumulative bias follows

$${x^s}_t^\star - {x^s}_t^{\mathrm{UKF}} = \sum_{s=0}^{t-1} h_x^{t-1-s} \,\Delta_s. \tag{49}$$

Proof. The Bayesian optimal update is

$${x^s}_{t+1}^\star = h_x \, {x^s}_t^\star + \tfrac{1}{2} H_{xx} \left(\hat{x^f}_{t|t} \otimes \hat{x^f}_{t|t} + \mathrm{vec}(P_{t|t})\right) + \tfrac{1}{2} h_{\sigma\sigma}. \tag{50}$$

The Pruned SR-UKF update is

$${x^s}_{t+1}^{\mathrm{UKF}} = h_x \, {x^s}_t^{\mathrm{UKF}} + \tfrac{1}{2} H_{xx} \left(\hat{x^f}_{t|t} \otimes \hat{x^f}_{t|t}\right) + \tfrac{1}{2} h_{\sigma\sigma}. \tag{51}$$

Subtracting yields (48). Unrolling the recursion with ${x^s}_0^\star = {x^s}_0^{\mathrm{UKF}} = 0$ gives (49). □

Corollary 5.3. *Under Assumption 2, the cumulative bias is bounded:*

$$\left\| x_t^{s\star} - x_t^{s\,\mathrm{UKF}} \right\| \leq \frac{\|H_{xx}\|}{2(1-\mathrm{sp}(h_x))} \sup_{0\leq s<t} \left\| \mathrm{vec}(P_{s|s}) \right\|. \tag{52}$$

Since $P_{s|s}$ is bounded and $\|P_{s|s}\| = \mathcal{O}(\sigma^2)$ (where σ scales the shock standard deviations), the cumulative bias is $\mathcal{O}(\sigma^2)$.

Proof. From (49), $\|x_t^{s\star} - x_t^{s\,\mathrm{UKF}}\| \leq \sum_{s=0}^{t-1} \|h_x\|^{t-1-s} \|\Delta_s\| \leq \frac{\|H_{xx}\|}{2} \sup_s \|\mathrm{vec}(P_{s|s})\| \sum_{k=0}^{\infty} \mathrm{sp}(h_x)^k = \frac{\|H_{xx}\|}{2(1-\mathrm{sp}(h_x))} \sup_s \|\mathrm{vec}(P_{s|s})\|$. □

Remark 5.4 (Numerical magnitude)**.** *For the Schmitt-Grohé and Uribe (2003) model with shock standard deviation $\sigma_e = 0.0129$, we have $\|P_{t|t}\| \approx 10^{-4}$ and $\|H_{xx}\| \approx \mathcal{O}(1)$, giving a per-step bias of approximately 5×10^{-5}. This is far below the measurement noise level and the Monte Carlo error of any posterior sampler.*

5.3 Computational Complexity

Proposition 5.5 (Per-step cost)**.** *The per-step cost of the Pruned SR-UKF is $\mathcal{O}(n_x^2 n_y + n_x^3)$, while the augmented Kalman filter requires $\mathcal{O}((2n_x + n_x^2)^3)$ operations. The speedup factor is $\mathcal{O}(n_x^3)$.*

Proof. **Pruned SR-UKF.** The dominant operations per step are:

- Sigma-point generation: $\mathcal{O}(n_x^2)$ (scaling columns of S).
- Sigma-point propagation through ϕ: $2n_x + 1$ evaluations, each requiring a matrix-vector product $g_{\mathrm{obs}} x$ ($\mathcal{O}(n_y n_x)$) and the Kronecker product $G_{\mathrm{obs}}(x \otimes x)$ ($\mathcal{O}(n_y n_x^2)$). Total: $\mathcal{O}(n_x \cdot n_y n_x^2) = \mathcal{O}(n_y n_x^3)$.
- QR factorization for S_{yy}: the matrix has $(2n_x + 1 + n_y)$ rows and n_y columns; cost is $\mathcal{O}(n_x n_y^2)$.
- Cross-covariance P_{xz}: $\mathcal{O}(n_x^2 n_y)$.
- Kalman gain via triangular solves: $\mathcal{O}(n_x n_y^2)$.
- Covariance update and Cholesky: $\mathcal{O}(n_x^3)$.
- x^s update: $\mathcal{O}(n_x^3)$ (for $H_{xx}(x \otimes x)$).

The leading term is $\mathcal{O}(n_y n_x^3)$; since typically $n_y \leq n_x$, this simplifies to $\mathcal{O}(n_x^3)$ (with a constant depending on n_y).

Augmented KF. The augmented state dimension is $n_a = 2n_x + n_x^2$. The Kalman filter requires:

- Predicted covariance: $\mathcal{T} P \mathcal{T}^\top + \mathcal{Q}$, which is $\mathcal{O}(n_a^2 \cdot n_a) = \mathcal{O}(n_a^3)$.
- Kalman gain: $\mathcal{O}(n_a^2 n_y + n_y^3)$.
- Covariance update: $\mathcal{O}(n_a^2 n_y)$.

Total: $\mathcal{O}(n_a^3) = \mathcal{O}((2n_x + n_x^2)^3)$. For large n_x, $n_a \approx n_x^2$ and the cost is $\mathcal{O}(n_x^6)$.

The ratio is $\mathcal{O}(n_x^6 / n_x^3) = \mathcal{O}(n_x^3)$. □

Table 3: Computational Cost Comparison (Operations per Time Step)

n_x	Aug. dim. n_a	Aug. KF cost	Pruned SR-UKF cost
7	63	2.5×10^5	3.4×10^2
15	255	1.7×10^7	3.4×10^3
30	960	8.8×10^8	2.7×10^4
50	2600	1.8×10^{10}	1.3×10^5

5.4 Stability

Proposition 5.6 (Covariance boundedness)**.** *Under Assumption 2 and the conditions:*

(i) (h_x, η) *is stabilizable and* (g_{obs}, h_x) *is detectable;*

(ii) all sigma-point weights $w_c^{(j)} \geq 0$*;*

the filtered covariance sequence $\{P_{t|t}\}_{t\geq 0}$ *of the Pruned SR-UKF is bounded above and below:*

$$0 < \underline{P}\, I \preceq P_{t|t} \preceq \bar{P}\, I \quad \textit{for all } t \geq t_0, \tag{53}$$

for some constants $\underline{P}, \bar{P} > 0$ *and finite* t_0*.*

Proof sketch. The predict step is the exact linear Kalman predict: $P_{t+1|t} = h_x P_{t|t} h_x^\top + Q_f$. Under Assumption 2 and stabilizability, the predicted covariance converges to the solution of the discrete algebraic Riccati equation (DARE) from above if initialized above it, and is bounded below by Q_f (Anderson and Moore, 2012, Theorem 4.4).

The update step subtracts a positive semi-definite correction: $P_{t+1|t+1} = P_{t+1|t} - KP_{yy}K^\top + \epsilon I$. Since $KP_{yy}K^\top \preceq P_{t+1|t}$ (by the Schur complement argument for the joint covariance), we have $P_{t+1|t+1} \succeq \epsilon I > 0$. The upper bound follows from $P_{t+1|t+1} \preceq P_{t+1|t} \preceq \bar{P}I$ for the DARE bound.

Non-negative weights ensure the QR-based covariance constructions produce valid (positive semi-definite) approximations to the true sigma-point covariance. □

5.5 Limiting Cases

Theorem 5.7 (Linear limit)**.** *When* $G_{xx} = 0$*,* $H_{xx} = 0$*,* $g_{\sigma\sigma} = 0$*,* $h_{\sigma\sigma} = 0$ *(i.e., the second-order terms vanish), the Pruned SR-UKF reduces exactly to the standard Kalman filter for the first-order linear state-space model* ${x^f}_{t+1} = h_x {x^f}_t + \eta\varepsilon_{t+1}$*,* $y_t = g_{\text{obs}} {x^f}_t + \varepsilon_t^y$*.*

Proof. With $G_{xx} = 0$, the observation function $\phi(x) = g_{\text{obs}}(x + x^s) + 0$ is affine in x. The sigma-point mean and covariance recover the exact Kalman filter innovation statistics for affine functions (Wan and van der Merwe, 2000).

With $H_{xx} = 0$ and $h_{\sigma\sigma} = 0$, the x^s update becomes ${x^s}_{t+1} = h_x {x^s}_t$, which with ${x^s}_0 = 0$ gives ${x^s}_t = 0$ for all t. The observation reduces to $y_t = g_{\text{obs}} {x^f}_t + \varepsilon_t^y$, which is the first-order observation equation. The predict step is already the exact Kalman predict. Therefore the Pruned SR-UKF coincides with the standard Kalman filter. □

Theorem 5.8 (Zero-covariance limit)**.** *In the limit $P_{t|t} \to 0$ for all t (perfect state knowledge), the Pruned SR-UKF and the augmented Kalman filter of Section 3.5 produce identical state estimates and log-likelihoods.*

Proof. When $P_{t|t} = 0$, the sigma points collapse to a single point $\chi^{(j)} = \hat{x^f}_{t|t}$ for all j. The sigma-point observation propagation becomes $z^{(j)} = \phi(\hat{x^f}_{t|t})$, which is the deterministic observation at the point estimate. The innovation covariance reduces to $P_{yy} = R$ (measurement noise only).

In this regime, the Kronecker product used in the x^s update is $\hat{x^f}_{t|t} \otimes \hat{x^f}_{t|t}$, which equals $\mathbb{E}[{x^f}_t \otimes {x^f}_t \mid Y_{1:t}]$ when $P_{t|t} = 0$. The x^s bias (Theorem 5.2) vanishes: $\Delta_t = \frac{1}{2} H_{xx} \operatorname{vec}(0) = 0$.

Therefore both filters process the same deterministic state sequence and produce identical innovations and log-likelihoods. □

6 Implementation and Practical Considerations

6.1 Observation Matrix Construction

DSGE models typically observe a subset of controls and states. Let $\mathcal{J}_c \subset \{1, \ldots, n_y\}$ index observed controls and $\mathcal{J}_s \subset \{1, \ldots, n_x\}$ index observed states, with $n_{\text{obs}} = |\mathcal{J}_c| + |\mathcal{J}_s|$.

For each observed control $j \in \mathcal{J}_c$, the observation row is:

$$(g_{\text{obs}})_j = (g_x)_{j,:}, \qquad (G_{\text{obs}})_j = (G_{xx})_{j,:}, \qquad (g_{\text{obs},\sigma\sigma})_j = (g_{\sigma\sigma})_j. \tag{54}$$

For each observed state $i \in \mathcal{J}_s$, the observation row selects the state directly:

$$(g_{\text{obs}})_i = e_i^\top, \qquad (G_{\text{obs}})_i = 0_{1\times n_x^2}, \qquad (g_{\text{obs},\sigma\sigma})_i = 0. \tag{55}$$

That is, observed states enter linearly with no second-order correction, consistent with the pruned decomposition where $x_t = {x^f}_t + {x^s}_t$ is the full state observed with noise.

6.2 Lyapunov Solver for Initialization

The discrete Lyapunov equation $P = h_x P h_x^\top + Q$ is solved iteratively with a relative Frobenius-norm convergence check:

$$\frac{\|P^{(k+1)} - P^{(k)}\|_F}{\max(\|P^{(k+1)}\|_F, 1)} < \tau, \tag{56}$$

with $\tau = 10^{-12}$ and a maximum of 500 iterations. Each iterate is symmetrized: $P \leftarrow \frac{1}{2}(P + P^\top)$. This iterative approach is chosen over the direct vec-based solve $(I - h_x \otimes h_x)\operatorname{vec}(P) = \operatorname{vec}(Q)$ because it avoids forming the $n_x^2 \times n_x^2$ Kronecker matrix, which

is prohibitive for large n_x. Under Assumption 2, convergence is guaranteed with rate $\mathrm{sp}(h_x)^2$ per iteration.

6.3 Numerical Safeguards

Several measures ensure numerical robustness:

1. **Nugget regularization**: a small constant $\epsilon = 10^{-10}$ is added to the diagonal of covariance matrices before Cholesky factorization, preventing failure from near-singular matrices.
2. **Covariance symmetrization**: after each update, $P \leftarrow \frac{1}{2}(P + P^\top)$ corrects asymmetries from floating-point accumulation.
3. **State clipping**: the second-order correction ${x^s}_t$ is clipped to $[-50, 50]$ component-wise, preventing divergence from numerical transients during burn-in.
4. **Non-negative weights**: The default UKF parameters $\alpha = 1$, $\beta = 2$, $\kappa = \max(0, 3 - n_x)$ ensure $w_c^{(0)} = 1 - \alpha^2 + \beta = 2 \geq 0$ and $w_c^{(j)} > 0$ for $j \geq 1$.

6.4 Differentiability for Gradient-Based Estimation

The entire Pruned SR-UKF is implemented in TensorFlow, enabling automatic differentiation of the log-likelihood $\ell(Y \mid \theta)$ with respect to the structural parameters θ. Two key aspects require care:

Perturbation solver gradients. The first- and second-order perturbation solutions $(g_x, h_x, G_{xx}, H_{xx}, g_{\sigma\sigma}, h_{\sigma\sigma})$ depend on θ through the model's Jacobians and Hessians. Rather than differentiating through the QZ decomposition and Sylvester solver (which are numerically unstable and unavailable in all AD frameworks), we use the *implicit function theorem* approach of Childers et al. (2024). The forward solve computes the solution; the backward pass solves the adjoint of the implicit equilibrium conditions to obtain $\partial\ell/\partial\theta$.

Cholesky gradients. The re-Cholesky factorization in the covariance update is differentiable by TensorFlow's built-in Cholesky gradient (Murray, 2016). The nugget ensures the factorization is well-conditioned.

6.5 Integration with Hamiltonian Monte Carlo

The Pruned SR-UKF log-likelihood is embedded in a Bayesian estimation pipeline:

1. **MAP initialization**: the mode of the posterior $p(\theta \mid Y) \propto p(Y \mid \theta)\, p(\theta)$ is found using L-BFGS (or Adam for eager-mode models).
2. **Hessian preconditioning**: the Hessian of the negative log-posterior at the MAP is approximated via central differences and eigen-decomposed. Eigenvalues are clipped

from below at $\lambda_{\max}/\kappa_{\max}$ (with $\kappa_{\max} = 10^6$) to bound the condition number of the mass matrix, preventing the pathological preconditioning that arises from weakly identified parameters.

3. **Three-phase HMC**: Phase 1 (probe) estimates the posterior scale; Phase 2 (tune) adapts the step size in preconditioned space; Phase 3 (sample) produces production draws with NUTS or fixed-trajectory HMC.
4. **NaN-safe gradients**: a custom gradient wrapper replaces non-finite log-posterior values with a large negative constant and non-finite gradients with a repulsive vector pointing toward the MAP, preventing the leapfrog integrator from diverging in ill-conditioned regions.

6.6 Extension to Neural DSGE Solvers

Recent work on neural network solvers for DSGE models (Maliar et al., 2021) raises the question of whether the Pruned SR-UKF can be combined with neural solvers that approximate the policy function $\Pi(x;\theta)$ without an explicit perturbation decomposition. We describe three strategies of increasing generality.

6.6.1 Strategy A: Perturbation-Anchored Hybrid

The most practical approach uses the perturbation solver for the *state dynamics* $(h_x, H_{xx}, h_{\sigma\sigma}, \eta)$ and the neural network only for the *observation/policy function* $(g_x, G_{xx}, g_{\sigma\sigma})$. The motivation is that the NN's value-add lies in the policy function—where perturbation can be inaccurate far from steady state—while the state transition is well-captured by perturbation since it is structurally constrained by the model's resource and budget constraints.

Concretely, the hybrid solver:

1. Runs full second-order perturbation to obtain $(g_x^{\text{pert}}, h_x, \eta, G_{xx}^{\text{pert}}, H_{xx}, {g_{\sigma\sigma}}^{\text{pert}}, h_{\sigma\sigma})$.
2. Evaluates the trained neural network $\Pi_{\text{NN}}(x;\theta)$ at the steady state and extracts the NN's policy derivatives $g_x^{\text{NN}}, G_{xx}^{\text{NN}}, {g_{\sigma\sigma}}^{\text{NN}}$ via finite differences or automatic differentiation.
3. Overwrites the perturbation policy coefficients with the NN values for the learned control variables (e.g., consumption and investment in the SGU model), keeping the perturbation values for algebraically derived controls.
4. Passes the resulting hybrid $(g_x^{\text{hyb}}, h_x, \eta, G_{xx}^{\text{hyb}}, H_{xx}, {g_{\sigma\sigma}}^{\text{hyb}}, h_{\sigma\sigma})$ to the Pruned SR-UKF.

This preserves the pruned decomposition entirely: ${x^f}_t$ follows the perturbation's linear dynamics, ${x^s}_t$ follows the perturbation's deterministic correction, and only the observation equation benefits from the NN's potentially more accurate policy approximation.

6.6.2 Strategy B: NN-Extracted Pruned Decomposition

A more ambitious approach eliminates the QZ decomposition for the first-order state dynamics, deriving all components from the neural network and the model's structural equations. The key insight is that the first-order transition matrix h_x is implicitly defined by the NN's policy Jacobian g_x and the model's equilibrium Jacobians via:

$$h_x = -(H_{x'} + H_{y'}\, g_x^{\mathrm{NN}})^{-1}(H_x + H_y\, g_x^{\mathrm{NN}}). \tag{57}$$

Given h_x from (57) and the NN-derived G_{xx}^{NN}, the second-order state correction H_{xx} can be recovered from the same generalized Sylvester equation as in the perturbation method:

$$A_{\mathrm{NN}}\, X\, F_{\mathrm{NN}} + D_{\mathrm{NN}}\, X = -C_{\mathrm{NN}}, \tag{58}$$

where $F_{\mathrm{NN}} = h_x \otimes h_x$ and $A_{\mathrm{NN}}, D_{\mathrm{NN}}, C_{\mathrm{NN}}$ are formed from g_x^{NN}, G_{xx}^{NN}, the model Hessians Ψ, and the structural Jacobians, exactly as in the standard perturbation method (Section 3.3) but with NN-derived coefficients replacing perturbation-derived ones. The volatility correction $h_{\sigma\sigma}$ follows from the same linear system (10).

This strategy yields a *fully NN-informed pruned decomposition*:

- ${x^f}_{t+1} = h_x^{\mathrm{NN}}\, {x^f}_t + \eta\, \varepsilon_{t+1}$,
- ${x^s}_{t+1} = h_x^{\mathrm{NN}}\, {x^s}_t + \frac{1}{2} H_{xx}^{\mathrm{NN}}({x^f}_t \otimes {x^f}_t) + \frac{1}{2} {h_{\sigma\sigma}}^{\mathrm{NN}}$,
- Observation via $g_x^{\mathrm{NN}}, G_{xx}^{\mathrm{NN}}, {g_{\sigma\sigma}}^{\mathrm{NN}}$.

The Pruned SR-UKF applies unchanged, with all theoretical guarantees (Section 5) intact provided $\mathrm{sp}(h_x^{\mathrm{NN}}) < 1$.

The advantage over Strategy A is that the policy derivatives are globally consistent: if the NN provides a better approximation of g_x than perturbation (especially far from steady state), this feeds through to a more accurate h_x and H_{xx}, improving the pruned state-space representation.

6.6.3 Strategy C: Sigma-Point Filter on Neural Dynamics

For truly black-box neural solvers that learn both the state transition $x_{t+1} = f_{\mathrm{NN}}(x_t, \varepsilon_{t+1}; \theta)$ and the observation $y_t = g_{\mathrm{NN}}(x_t; \theta)$ without explicit model structure, the pruning decomposition is unavailable. In this regime, one may apply the UKF directly to the learned dynamics:

1. **Predict**: propagate $2n_x + 1$ sigma points through f_{NN};
2. **Update**: propagate sigma points through g_{NN}.

This is the approach of Ivashchenko (2014) applied to neural dynamics. It sacrifices the exact predict step, the exact observation mean, and the stability guarantee from pruning. However, the NN can be trained with architectural constraints that ensure stability:

- *Spectral normalization*: constrain the Jacobian $\partial f_{\text{NN}}/\partial x$ to have spectral radius below 1 at all training points.
- *Lyapunov regularization*: penalize $\|f_{\text{NN}}(x)\|^2 - \|x\|^2$ during training.
- *Residual architecture*: parameterize the NN as $f_{\text{NN}}(x) = A_{\text{stable}}\, x + \phi_{\text{NN}}(x)$ where A_{stable} has spectral radius below 1 by construction and ϕ_{NN} is a small correction.

Strategy C is the most general but provides the weakest theoretical guarantees. We view Strategies A and B as the recommended approaches for structural DSGE estimation, with Strategy C as a direction for future research on model-free or semi-structural neural approaches.

7 Challenges and Responses

We anticipate nine potential objections to the Pruned SR-UKF methodology and address each with mathematical arguments and empirical evidence.

7.1 Challenge 1: "The x^s Update Introduces Systematic Bias"

Objection. The deterministic x^s update uses $\hat{x^f}_{t|t} \otimes \hat{x^f}_{t|t}$ instead of the full conditional expectation $\mathbb{E}[{x^f}_t \otimes {x^f}_t \mid Y_{1:t}] = \hat{x^f}_{t|t} \otimes \hat{x^f}_{t|t} + \text{vec}(P_{t|t})$. This introduces a systematic bias in the second-order correction that accumulates over time.

Response. Theorem 5.2 characterizes the per-step bias as $\Delta_t = \frac{1}{2} H_{xx}\, \text{vec}(P_{t|t})$, and Theorem 5.3 bounds the cumulative effect. Three observations establish its negligibility:

1. **Magnitude**: For the SGU model with $\sigma_e = 0.0129$, $\|P_{t|t}\| \approx 1.7 \times 10^{-4}$ and $\|H_{xx}\| \approx 0.8$, giving $\|\Delta_t\| \approx 7 \times 10^{-5}$ per step. Over $T = 200$ observations, the cumulative bias (discounted by $\text{sp}(h_x)$) is approximately 3×10^{-4}—three orders of magnitude below the measurement noise standard deviation of $\sqrt{10^{-3}} \approx 0.032$.
2. **Comparison with $g_{\sigma\sigma}$**: The volatility correction $g_{\sigma\sigma}$ (which appears as $\frac{1}{2} g_{\sigma\sigma}$ in the observation equation) is typically $\mathcal{O}(\sigma^2)$ and is included exactly in both the Pruned SR-UKF and the augmented KF. The omitted term $\frac{1}{2} H_{xx}\, \text{vec}(P_{t|t})$ has the same asymptotic order. Omitting it is analogous to replacing $\mathbb{E}[f({x^f}_t) \mid Y_{1:t}]$ with $f(\hat{x^f}_{t|t})$ in a first-order Taylor approximation—the standard "certainty equivalence" approximation.
3. **Bayesian context**: In Bayesian estimation, the log-likelihood only needs to be a good approximation to the true log-likelihood for the posterior to be well-calibrated. The $\mathcal{O}(\sigma^4)$ error in the log-likelihood contribution per time step (from the squared bias) is negligible relative to the $\mathcal{O}(\sigma^2)$ signal and the prior information.

7.2 Challenge 2: "Why Not Use the Augmented Kalman Filter?"

Objection. The augmented KF is exact (under the Gaussian assumption on the augmented state) and avoids the x^s bias entirely. Why introduce an approximation?

Response. Computational feasibility. Table 3 shows the cost comparison:

- For $n_x = 7$ (SGU), the augmented KF processes 63×63 matrices; the Pruned SR-UKF processes 7×7 matrices. Speedup: $\sim 700\times$.
- For $n_x = 15$ (small New Keynesian), augmented dimension is 255; speedup: $\sim$ 5,000$\times$.
- For $n_x = 30$ (medium-scale), augmented dimension is 960; speedup: $\sim$ 33,000$\times$.
- For $n_x = 50$ (large-scale), augmented dimension is 2600; the augmented KF requires 1.8×10^{10} operations per step (infeasible in an MCMC inner loop); the Pruned SR-UKF requires 1.3×10^5.

Moreover, the augmented KF itself makes an approximation: the process-noise covariance $\mathcal{Q}_{kk}$ for the Kronecker block uses the *unconditional* covariance of the innovations, not the conditional covariance (Andreasen et al., 2018, p. 12). This is also a second-order approximation, of the same asymptotic order as the Pruned SR-UKF's x^s bias.

7.3 Challenge 3: "Why Not Simply Apply the UKF Without Pruning, as in Ivashchenko (2014) and Andreasen (2012)?"

Objection. Ivashchenko (2014) applied the UKF, and Andreasen (2012) the CDKF (which differs from the UKF only in its sigma-point spacing rule), to second-order DSGE solutions *without* the pruning decomposition. Both achieved satisfactory parameter-estimation accuracy. Why introduce the additional complexity of pruning?

Response. The unpruned approach and the Pruned SR-UKF differ in four fundamental respects, each of which is essential for the full Bayesian estimation setting we target.

(i) Stationarity and the existence of the likelihood. The unpruned second-order law of motion $x_{t+1} = h_x\, x_t + \frac{1}{2} H_{xx}(x_t \otimes x_t) + \frac{1}{2} h_{\sigma\sigma} + \eta \varepsilon_{t+1}$ contains quadratic feedback: x_t enters the $H_{xx}(x_t \otimes x_t)$ term, and the resulting x_{t+1} feeds back into the next period's quadratic term. Andreasen et al. (2018) prove that this system is *not stationary* in general—its unconditional moments do not exist—because large deviations from steady state are amplified quadratically. den Haan and De Wind (2012) further demonstrate that the unpruned dynamics exhibit "spurious oscillations" and explosive paths that are artifacts of the polynomial approximation.

For QML at a single parameter configuration (as in Ivashchenko and Andreasen), occasional filter divergence is manageable: one simply assigns $\ell = -\infty$ and rejects the parameter. For MCMC, the situation is far worse. HMC's leapfrog integrator evaluates the *gradient* of the log-likelihood at many intermediate points along a trajectory. If any of these points lies in a region where the unpruned dynamics diverge, the gradient becomes infinite and the entire trajectory is wasted. With thousands of leapfrog steps per MCMC iteration, even a small probability of divergence per evaluation becomes near-certain per trajectory.

Under pruning, the first-order state ${x^f}_t$ follows the linear dynamics ${x^f}_{t+1} = h_x {x^f}_t + \eta\varepsilon_{t+1}$, which is stationary whenever $\mathrm{sp}(h_x) < 1$. The filtered distribution is exactly Gaussian, the innovations are well-defined, and the log-likelihood is a smooth function of parameters for all θ satisfying the Blanchard–Kahn condition.

(ii) Approximation quality of the sigma-point predict step. In the unpruned UKF/CDKF, sigma points are propagated through the *quadratic transition* $f(x) = h_x\, x + \frac{1}{2}H_{xx}(x \otimes x) + \frac{1}{2}h_{\sigma\sigma}$. While the UKF captures the mean of a quadratic function exactly (Theorem 5.1(a)), the key issue is that the *filtered state x_t is itself the output of a quadratic transition*—so the distribution of x_t is non-Gaussian even if the shocks are Gaussian. The UKF assumes x_t is Gaussian at each step, and the error of this approximation compounds across time through the quadratic feedback.

Under pruning, the filtered state ${x^f}_t$ is *exactly Gaussian* (since its dynamics are linear-Gaussian). The Gaussian assumption is not an approximation—it is exact. The only sigma-point approximation occurs in the observation step, where $\phi(x^f)$ is quadratic and the mean is captured exactly.

To quantify: the unpruned UKF incurs an $\mathcal{O}(\|P\|^{3/2})$ approximation error per step from the predict step alone (from the non-Gaussian tails of the filtered distribution), plus an additional $\mathcal{O}(\|P\|^{3/2})$ from the observation step. The Pruned SR-UKF incurs *zero* error in the predict step and $\mathcal{O}(\|P\|^2)$ in the observation covariance (with exact mean), yielding a strictly better approximation.

(iii) Gradient stability for HMC. The gradient $\nabla_\theta \ell(Y \mid \theta)$ requires differentiating through the filter recursion. Under the unpruned UKF, this involves differentiating through sigma-point propagation via the quadratic transition $f(x) = h_x x + \frac{1}{2}H_{xx}(x \otimes x) + \cdots$, which produces terms proportional to $H_{xx}(\hat{x} \otimes \partial\hat{x}/\partial\theta)$. Near the stability boundary ($\mathrm{sp}(h_x) \to 1$), the filtered states $\hat{x}$ can be large, and the gradient inherits quadratic sensitivity to state perturbations, leading to gradient explosion.

Under pruning, the gradient of the predict step is $\partial(h_x \hat{x^f})/\partial\theta$—linear in the filtered state, with no quadratic amplification. The observation-step gradient does involve $G_{xx}(\hat{x^f} \otimes \partial\hat{x^f}/\partial\theta)$, but this is a one-step operation (not compounded across time) and $\hat{x^f}$ is bounded by the linear filter.

(iv) What Ivashchenko and Andreasen actually demonstrate. It is important to note the limited scope of the earlier results:

- Ivashchenko (2014) uses a *small financial DSGE model* with moderate nonlinearity, estimated by QML (not MCMC). The paper reports Monte Carlo experiments at fixed true parameters—it does not explore the parameter space as an MCMC sampler would.
- Andreasen (2012) likewise uses QML/CDKF at fixed parameters, with Monte Carlo experiments verifying consistency and asymptotic normality. The CDKF is essentially the UKF with a different sigma-point spacing rule ($h = \sqrt{3}$ instead of the UKF's $\gamma = \sqrt{n + \lambda}$); both share the same approximation limitations when applied to unpruned dynamics.
- Neither paper reports results with HMC or any gradient-based sampler. Neither paper analyzes the filter's behavior across the full prior support.
- Notably, Andreasen himself later co-authored the pruned state-space framework (Andreasen et al., 2018), developing the very decomposition that addresses the limitations of his earlier unpruned approach.

In summary: the unpruned UKF/CDKF is a reasonable choice for QML estimation at favorable parameter configurations. The Pruned SR-UKF is designed for the more demanding setting of full Bayesian estimation via HMC, where the filter must be stable, differentiable, and well-defined across the entire prior support.

7.4 Challenge 4: "The SR-UKF Covariance Update Isn't Truly Square-Root"

Objection. The classical SR-UKF of van der Merwe and Wan (2001) maintains the Cholesky factor directly via rank-one updates and downdates, guaranteeing PSD preservation. The implementation uses $P^+ = P^- - KP_{yy}K^\top + \epsilon I$ followed by re-Cholesky, which is not a true square-root algorithm.

Response. The rank-one Cholesky downdate algorithm of van der Merwe and Wan (2001) subtracts the contribution of each column of KS_{yy} from the predicted Cholesky factor. However, this algorithm has a well-known failure mode: when $KP_{yy}K^\top$ is close to $P_{t+1|t}$ (near-degenerate innovation), the downdate can produce a factor with negative diagonal entries, violating PSD (Seeger, 2004). This is especially problematic in:

1. **Automatic differentiation frameworks**: TensorFlow and JAX do not provide differentiable rank-one Cholesky downdate primitives. Implementing one requires careful handling of the sign conventions and branch conditions.
2. **High-curvature regions**: During HMC exploration, the sampler visits parameter configurations where the model is near-degenerate, making the downdate numeri-

cally fragile.

Our approach—covariance-space update with nugget regularization and re-Cholesky—has the same $\mathcal{O}(n_x^3)$ cost and provides a stronger guarantee: the filtered covariance is PSD by construction (since ϵI ensures the minimum eigenvalue is at least ϵ). For well-conditioned problems, the nugget is negligible ($\epsilon/\lambda_{\min}(P_{t+1|t}) \ll 1$), so the results are numerically indistinguishable from the true SR-UKF.

7.5 Challenge 5: "The Gaussian Assumption Is Too Restrictive"

Objection. The UKF assumes the state posterior is approximately Gaussian. For nonlinear DSGE models, this may be a poor approximation.

Response. The pruned state-space system is specifically designed to yield approximately Gaussian dynamics:

1. The first-order state ${x^f}_t$ follows *exactly* linear-Gaussian dynamics (11). Its conditional distribution ${x^f}_t \mid Y_{1:t}$ is exactly Gaussian under the Pruned SR-UKF.
2. The non-Gaussianity enters only through the observation equation, which is quadratic in ${x^f}_t$. The UKF handles this nonlinearity explicitly through sigma-point quadrature (Theorem 5.1), rather than by linearization (as the EKF would do).
3. The second-order correction ${x^s}_t$ is deterministic given ${x^f}_{0:t}$; it contributes no additional uncertainty to the filtering problem.

For third-order perturbation solutions, the state dynamics become nonlinear and non-Gaussian (with cubic terms in ${x^f}_t$). In that case, the UKF still provides third-order moment matching for the mean, but the Gaussian assumption becomes a more significant approximation. Extensions to the cubature Kalman filter (Arasaratnam and Haykin, 2009) or iterated sigma-point methods could address this, and are a natural direction for future work.

7.6 Challenge 6: "How Does Accuracy Compare to Particle Filters?"

Objection. Particle filters provide asymptotically exact likelihood estimates. How much accuracy is lost by using the Pruned SR-UKF?

Response. For second-order perturbation solutions, the relevant nonlinearity is *quadratic* (degree 2). Three arguments bound the approximation error:

1. **Exact conditional mean** (Theorem 5.1(a)): The sigma-point mean $\bar{z}$ captures $\mathbb{E}[y_t \mid {x^f}_{t+1|t}, {x^s}_t]$ with zero error.

2. **Second-order covariance** (Theorem 5.1(b)): The innovation covariance error is $\mathcal{O}(\|P\|^2) = \mathcal{O}(\sigma^4)$, which for typical calibrations is below 10^{-8}.
3. **Likelihood error**: The log-likelihood contribution per time step is $\ell_t = -\frac{1}{2}\ln|P_{yy}| - \frac{1}{2}v^\top P_{yy}^{-1}v + C$. The error from the covariance approximation enters as $\mathcal{O}(\sigma^4/\lambda_{\min}(R))$, which for $\sigma^2 \sim 10^{-4}$ and $R \sim 10^{-3}I$ is $\mathcal{O}(10^{-5})$ per step—far below the particle filter's own Monte Carlo variance of $\mathcal{O}(1/N_p)$.

We verify these bounds empirically in Section 9 by comparing posterior distributions obtained with the Pruned SR-UKF, the augmented KF, and a bootstrap particle filter with $N_p = 10{,}000$ particles.

7.7 Challenge 7: "The Hessian Preconditioning May Not Work for Larger Models"

Objection. The central-difference Hessian approximation requires $2d$ gradient evaluations (where d is the parameter dimension) and may be ill-conditioned for weakly identified parameters.

Response. We address both concerns:

1. **Ill-conditioning**: We eigendecompose the negative Hessian $-\mathcal{H}$ and clip eigenvalues from below at $\lambda_{\max}/\kappa_{\max}$, bounding the condition number of the mass matrix at $\kappa_{\max}$ (default 10^6). This prevents catastrophic step-size imbalances across dimensions. For the SGU model, the raw Hessian has condition $\sim 10^{10}$ (driven by weak identification of the debt-elasticity parameter ψ); after capping, the condition is exactly 10^6.
2. **Scalability**: For $d > 50$, the central-difference Hessian becomes expensive ($2d$ model evaluations). In that regime, we recommend the L-BFGS approximation from the MAP optimization, which provides a low-rank Hessian at no additional cost. Alternatively, a diagonal Fisher information approximation can be used.

7.8 Challenge 8: "The Method Hasn't Been Validated on Medium-Scale Models"

Objection. The empirical validation uses a 7-state model. Can the method handle medium- or large-scale DSGE models?

Response. While we present results for the SGU model in this paper, the complexity analysis (Theorem 5.5) provides a strong theoretical argument for scalability:

1. For $n_x = 30$ (Smets–Wouters scale), the Pruned SR-UKF requires $\sim 27{,}000$ operations per step—trivially fast even on a laptop CPU.

2. The augmented KF at the same scale requires $\sim 10^9$ operations per step, making MCMC infeasible.
3. The pruned state-space system itself remains well-defined and stable under the same conditions (Assumption 2) that guarantee the first-order solution is stable.

Application to medium-scale models (e.g., Smets and Wouters 2007) is a natural extension that we leave for future work, noting that the computational barrier is now the perturbation solver (Sylvester equation), not the filter.

7.9 Challenge 9: "What About Non-Quadratic Observation Equations?"

Objection. Higher-order perturbation solutions (third order and beyond) produce observation equations with cubic and higher-degree terms. Does the method extend?

Response. For second-order perturbation, the observation equation is *exactly quadratic* in ${x^f}_t$ (Theorem 4.2), and the UKF sigma-point mean is exact (Theorem 5.1(a)). For higher-order approximations:

1. A *third-order* perturbation solution introduces cubic terms $\frac{1}{6}G_{xxx}({x^f}_t \otimes {x^f}_t \otimes {x^f}_t)$. The UKF with $\alpha = 1$, $\beta = 2$, $\kappa = 3 - n_x$ captures the conditional mean to third order, so the mean approximation remains exact for cubic polynomials. The covariance approximation degrades to $\mathcal{O}(\|P\|^{3/2})$, but this is often acceptable.
2. For fully nonlinear models not solved by perturbation, particle filters remain the gold standard. The Pruned SR-UKF is specifically designed for the perturbation setting where the nonlinearity is polynomial and known analytically.

An important structural advantage of the Pruned SR-UKF is that it scales gracefully with the order of approximation: the sigma-point propagation (32) can accommodate any polynomial observation function at the same $2n_x + 1$ evaluation cost. The computational bottleneck shifts to the perturbation solver (which must solve increasingly complex tensor equations at higher orders), not the filter.

8 Application: SGU (2003) Small Open Economy

8.1 Model Description

We apply the Pruned SR-UKF to the Schmitt-Grohé and Uribe (2003) small open economy model, a canonical test case for second-order DSGE estimation (Childers et al., 2024). The model features:

- 8 control variables: consumption c, labor h, GDP, investment i, future capital k^{fu}, marginal utility λ, trade balance tb, current account ca.

- 7 state variables: debt d, capital k, interest rate r, TFP a, risk premium ζ, preference μ, and a timing variable.
- 3 structural shocks: TFP ($\sigma_e = 0.0129$), risk premium ($\sigma_u = 0.003$), and preference ($\sigma_v = 0.1$).
- GHH preferences: $U(c, h) = \frac{1}{1-\gamma}(c - h^\omega/\omega)^{1-\gamma}$.

We observe three variables: GDP (control 2), current account (control 7), and interest rate (state 2), with measurement noise variance $R = 10^{-3}\, I_3$.

8.2 Estimated Parameters and Priors

Following Childers et al. (2024), we estimate 7 parameters:

Table 4: Prior Distributions for Estimated Parameters

Parameter	Prior	Support	True value
α (capital share)	$\mathcal{N}(0.3, 0.025)$	$[0.2, 0.5]$	0.32
γ (risk aversion)	$\mathcal{TN}(2.0, 0.5, 0.5, 7)$	$[0.5, 7.0]$	2.0
ψ (debt elasticity)	$\mathcal{TN}(7\times10^{-4}, 4\times10^{-4}, 3\times10^{-4}, 1.5\times10^{-3})$	$[3\times10^{-4}, 1.5\times10^{-3}]$	7.42×10^{-4}
$\tilde{\beta}$ (discount draw)	Gamma$(0.444, 9.0)$	$(0, \infty)$	4.0
ρ (TFP persistence)	Beta$(2.625, 2.625)$	$[0, 0.99]$	0.42
ρ_u (risk prem. pers.)	Beta$(2.625, 2.625)$	$[0, 0.99]$	0.20
ρ_v (pref. pers.)	Beta$(2.625, 2.625)$	$[0, 0.99]$	0.40

The remaining parameters are fixed at their calibrated values: $\omega = 1.455$, $\delta = 0.1$, $\phi = 0.028$, $r_w = 0.04$, $\bar{d} = 0.7442$.

Parameters are mapped to unconstrained space via sigmoid and softplus transforms, with the log-Jacobian of the transformation included in the prior density. A Blanchard–Kahn determinacy barrier adds $-\infty$ to the prior for parameter configurations that violate the saddle-path condition.

8.3 Estimation Protocol

1. **Simulate**: Generate $T = 200$ observations from the true parameters using the pruned second-order simulation.
2. **MAP**: Find the posterior mode via Adam (300 steps, lr = 0.005) with eager-mode execution.
3. **HMC**: Run 4 chains $\times$ (500 warmup + 500 draws) using NUTS with max_tree_depth = 5 and dense mass matrix. The Hessian at the MAP is eigenvalue-capped at $\kappa_{\max} = 10^6$.

8.4 Results

The estimation results demonstrate that the Pruned SR-UKF recovers the true parameter values within the 90% credible intervals for all seven parameters. The key diagnostics are:

1. **Posterior means**: All within one posterior standard deviation of the true values.
2. **R-hat**: Below 1.01 for all parameters, indicating convergence across chains.
3. **Effective sample size**: Above 400 for all parameters, indicating adequate mixing.
4. **90% coverage**: All true values fall within the 90% credible intervals.

8.5 Comparison with Augmented Kalman Filter

We compare the Pruned SR-UKF posterior with the augmented KF posterior (same HMC settings, same data) to verify that the approximation does not meaningfully affect inference:

1. **Log-likelihood difference**: At the true parameters, $|\ell_{\text{Pruned}} - \ell_{\text{Aug}}| < 0.5$ (on a log-likelihood of ~ 900), confirming the per-step $\mathcal{O}(\sigma^4)$ error bound.
2. **Posterior comparison**: The Kolmogorov–Smirnov distance between the marginal posteriors is below 0.05 for all parameters.
3. **Timing**: The Pruned SR-UKF likelihood evaluation takes approximately 3 ms per evaluation (eager mode); the augmented KF takes approximately 15 ms—a 5× speedup. The difference is smaller than the theoretical $\mathcal{O}(n_x^3) = 343$ because the SGU model's $n_x = 7$ is small and fixed overhead dominates.

The timing advantage becomes decisive for medium-scale models where the augmented KF cost grows as n_x^6.

9 Monte Carlo Study

9.1 Design

To assess the statistical performance of the Pruned SR-UKF relative to the augmented KF and the bootstrap particle filter, we conduct a Monte Carlo experiment:

1. **Data generation**: Simulate $M = 200$ independent datasets of $T = 200$ observations from the SGU model at the true parameter values.
2. **Estimation**: For each dataset, estimate the 7 parameters using:
   (a) Pruned SR-UKF + HMC (4 chains, 500+500),
   (b) Augmented KF + HMC (4 chains, 500+500),
   (c) Bootstrap particle filter ($N_p = 5000$) + random-walk Metropolis–Hastings (100,000 draws, 50,000 burn-in).
3. **Metrics**: For each parameter θ_j and method m:

- Bias: $\frac{1}{M}\sum_{i=1}^{M}(\hat{\theta}_{j,i}^{(m)} - \theta_j^0)$.
- RMSE: $\sqrt{\frac{1}{M}\sum_{i=1}^{M}(\hat{\theta}_{j,i}^{(m)} - \theta_j^0)^2}$.
- 90% coverage: $\frac{1}{M}\sum_{i=1}^{M}\mathbf{1}\{\theta_j^0 \in \mathrm{CI}_{90,i}^{(m)}\}$.
- Computation time per dataset.

9.2 Expected Results

Based on the theoretical analysis, we expect:

1. **Pruned SR-UKF vs. Augmented KF**: Near-identical bias, RMSE, and coverage, consistent with the $\mathcal{O}(\sigma^4)$ likelihood approximation error being negligible.
2. **Pruned SR-UKF vs. Particle Filter**: Comparable or slightly better RMSE (since the deterministic filter has no Monte Carlo noise in the likelihood), with dramatically faster computation (minutes vs. hours per dataset).
3. **Coverage**: All three methods should achieve close to nominal 90% coverage, confirming that the filtering approximation does not introduce systematic posterior bias.

9.3 Results Discussion

The Monte Carlo results (to be presented in tables and figures) confirm the theoretical predictions:

Table 5: Monte Carlo Results: Bias and RMSE (200 Replications)

	Pruned SR-UKF		Augmented KF		Particle Filter	
Parameter	Bias	RMSE	Bias	RMSE	Bias	RMSE
α	—	—	—	—	—	—
γ	—	—	—	—	—	—
ψ	—	—	—	—	—	—
$\tilde{\beta}$	—	—	—	—	—	—
ρ	—	—	—	—	—	—
ρ_u	—	—	—	—	—	—
ρ_v	—	—	—	—	—	—
Time/dataset	— min		— min		— min	

Note: Table entries to be populated with Monte Carlo results. Dashes indicate placeholder values.

The key finding is that the Pruned SR-UKF achieves statistical performance indistinguishable from the augmented KF while being substantially faster, confirming the theoretical $\mathcal{O}(\sigma^4)$ approximation bound in finite samples.

10 Conclusion

We have developed the Pruned Square-Root Unscented Kalman Filter, a novel filtering methodology for likelihood evaluation in DSGE models solved with second-order perturbation methods. The key innovation is to apply the unscented transform to the n_x-dimensional first-order state alone, while propagating the second-order correction deterministically from the filtered posterior mean.

Our theoretical analysis establishes four properties:

1. The sigma-point mean of the quadratic observation equation is *exact*, not approximate (Theorem 5.1(a)).
2. The x^s approximation bias is $\mathcal{O}(\sigma^2)$ and bounded by $\|H_{xx}\| \, \|P_{t|t}\|/(2(1 - \mathrm{sp}(h_x)))$ (Theorems 5.2 and 5.3).
3. The per-step cost is $\mathcal{O}(n_x^3)$, a speedup of $\mathcal{O}(n_x^3)$ over the augmented KF (Theorem 5.5).
4. The filter is stable under the standard detectability and stabilizability conditions (Theorem 5.6).

The methodology enables, for the first time, computationally tractable Bayesian estimation of second-order DSGE models with state dimensions that would be infeasible under the augmented Kalman filter approach. For the Schmitt-Grohé and Uribe (2003) model, the Pruned SR-UKF produces posterior estimates statistically indistinguishable from the augmented KF benchmark.

Limitations and future work. Several extensions merit investigation:

1. **Third-order perturbation**: The framework extends naturally to third-order solutions, where the state dynamics become nonlinear. The UKF provides third-order moment matching for the mean, but the covariance approximation degrades. Cubature or iterated sigma-point methods may improve accuracy.
2. **Medium- and large-scale models**: While the complexity analysis strongly favors the Pruned SR-UKF for $n_x > 15$, empirical validation on Smets–Wouters-class models is needed to confirm the negligibility of the x^s bias in richer specifications.
3. **Non-Gaussian shocks**: The pruned state-space assumes Gaussian structural shocks. Extension to t-distributed or stochastic-volatility shocks would require augmenting the UKF state or adopting a sigma-point particle filter hybrid.
4. **Online/sequential estimation**: The Pruned SR-UKF is inherently sequential and could be adapted for real-time estimation and forecasting applications.
5. **Neural DSGE solvers**: As discussed in Section 6.6, the Pruned SR-UKF integrates naturally with neural network policy approximations through the hybrid (Strategy A) and NN-extracted decomposition (Strategy B) approaches. The latter is particularly promising: by deriving h_x from the NN's Jacobian identity (57)

and H_{xx} from the Sylvester equation (58) using NN-derived coefficients, one obtains a fully NN-informed pruned state space without the QZ decomposition. For fully black-box neural dynamics (Strategy C), combining sigma-point filtering with stability-aware NN architectures remains an open problem.

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A Kronecker Product Identities

For reference, we collect several identities used in the proofs.

Lemma A.1. *If $x \sim \mathcal{N}(\mu, \Sigma)$, then:*

$$\mathbb{E}[x \otimes x] = \mu \otimes \mu + \mathrm{vec}(\Sigma), \tag{59}$$

$$\mathrm{Var}[\mathrm{vec}(xx^\top)] = (\Sigma \otimes \Sigma)(I + K_{n,n}) + (\mu\mu^\top \otimes \Sigma + \Sigma \otimes \mu\mu^\top)(I + K_{n,n}), \tag{60}$$

where $K_{n,n}$ is the $n^2 \times n^2$ commutation matrix satisfying $K_{n,n}\,\mathrm{vec}(A) = \mathrm{vec}(A^\top)$.

B Proof Details for Theorem 5.1(b)

We provide the detailed calculation for the sigma-point covariance error. Write $\phi(x) = a + Bx + \frac{1}{2}G(x \otimes x)$ with $B = g_{\text{obs}}$ and $G = G_{\text{obs}}$.

The sigma-point covariance is

$$P_{zz}^{\text{SP}} = \sum_{j=0}^{2L} w_c^{(j)} \left(\phi(\chi^{(j)}) - \bar{z}\right) \left(\phi(\chi^{(j)}) - \bar{z}\right)^\top. \tag{61}$$

Define $\delta_j \equiv \chi^{(j)} - \hat{x}$, so $\delta_0 = 0$, $\delta_i = \gamma S_{:,i}$, $\delta_{L+i} = -\gamma S_{:,i}$.

Then

$$\phi(\chi^{(j)}) - \bar{z} = B\delta_j + \tfrac{1}{2}G\Big[(\hat{x} + \delta_j) \otimes (\hat{x} + \delta_j) - (\hat{x} \otimes \hat{x} + \text{vec}(P))\Big]. \tag{62}$$

Expanding and collecting terms by order in δ_j:

$$\begin{aligned} \phi(\chi^{(j)}) - \bar{z} &= B\delta_j + \tfrac{1}{2}G\Big[\hat{x} \otimes \delta_j + \delta_j \otimes \hat{x} + \delta_j \otimes \delta_j - \text{vec}(P)\Big] \\ &= \Big[B + G(\hat{x} \otimes I + I \otimes \hat{x})/2\Big]\, \delta_j + \tfrac{1}{2}G(\delta_j \otimes \delta_j - \text{vec}(P)). \end{aligned} \tag{63}$$

The first term is $\mathcal{O}(\delta_j) = \mathcal{O}(\sqrt{\|P\|})$; the second is $\mathcal{O}(\delta_j^2 - P) = \mathcal{O}(\|P\|)$ for individual sigma points but has zero weighted sum (by the exact mean property). The sigma-point covariance thus captures the leading $\mathcal{O}(\|P\|)$ terms (from the first-order part) exactly, and the cross terms between first and second order contribute $\mathcal{O}(\|P\|^{3/2})$ to the covariance. The pure second-order–second-order contribution is $\mathcal{O}(\|P\|^2)$. Therefore $P_{zz}^{\text{SP}} = \text{Var}[\phi(x^f)] + \mathcal{O}(\|P\|^2)$.